Explaining the Historical Rise and Recent Decline in Social Security Disability Insurance Enrollment

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Abstract

After substantial growth in the 1990s and 2000s, enrollment in the U.S. Social Security Disability Insurance (SSDI) program has been declining since 2013. We use detailed administrative data to quantify the contributions of various factors to trends in SSDI enrollment, focusing on the decline in the 2010s. A statistical decomposition suggests that the majority of the decline in SSDI enrollment since 2013 is attributable to declines in application rates—and, to a lesser extent, award rates—within demographic groups. There is very little contribution from changes over time in demographic characteristics, eligibility, or exit from SSDI. The decline in SSDI enrollment rates is disproportionately driven by low-to-middle-skilled men who, over time, have become less likely to apply for SSDI and more likely to work. Consistent with this descriptive evidence, we present results from a causal analysis suggesting that improved labor market opportunity for less-skilled men is a key explanation for the decline in SSDI applications. We also investigate several other popular hypotheses for the decline in SSDI applications, including lower award rates at the appeals level, and find evidence at odds with them.

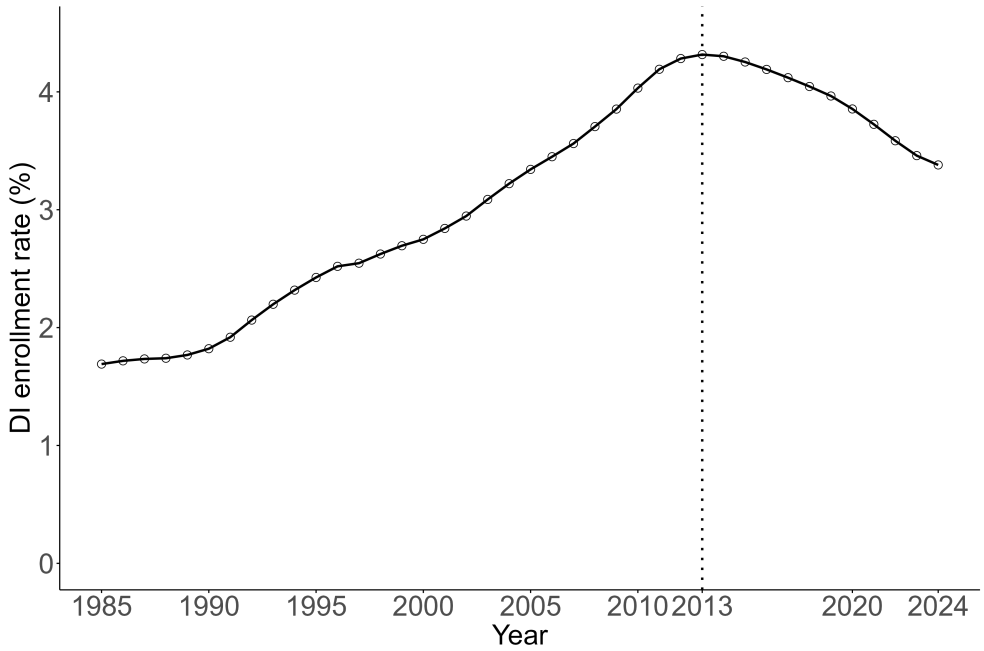
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The growth of the U.S. Social Security Disability Insurance (SSDI) program in the 1990s and 2000s gained widespread attention from policymakers, researchers, and the public, fueling concerns about an impending fiscal crisis and about misaligned incentives for Americans who are potentially capable of working. The growth was attributed to a combination of population aging, expansion of medical eligibility, and high benefit amounts relative to the potential earnings of low-skilled workers (Autor and Duggan, 2003, 2006; Reno, 2011; Congressional Budget Office, 2012; Liebman, 2015).

Figure 1: SSDI enrollment has been falling since 2013



Notes: SSDI enrollment rate is number of SSDI recipients divided by population aged 18 to 66 years.
Sources: Social Security Administration (2024) (Table 49) and Social Security Administration (2023).
See `https://www.ssa.gov/oact/HistEst/Population/2023/Population2023.html`.

Contrary to many projections from that time, enrollment in the SSDI program has actually been declining over the past decade, as shown in Figure 1. While other developed countries, including the United Kingdom, Australia, Sweden, and the Netherlands, have also experienced declines in disability enrollment, those declines are likely related to policy reforms that were intended to reduce enrollment (McVicar, Wilkins and Ziebarth, 2023). In contrast, there have been no explicit reforms in the U.S. to reduce SSDI enrollment. The reasons for the decline in SSDI, the future trajectory of the program, and the normative implications of the decline remain a mystery.

Of course, it is not surprising that SSDI *application* rates declined during the immediate recovery from the Great Recession. Maestas, Mullen and Strand (2021) show that the Great

Recession induced one million additional SSDI applications relative to normal times. The application rate began falling in 2010 as economic conditions improved. What is more surprising is that the application rate continued to decline into the late 2010s and early 2020s, and that the enrollment rate itself began declining in 2013.

Several explanations for the SSDI enrollment decline have been suggested. Perhaps the leading hypothesis is that the retraining of administrative law judges reduced award rates (Hoynes, Maestas and Strand, 2024) and may have additionally discouraged potential applicants from applying (Ruffing, 2020).[1] Another policy-related explanation is a higher burden of applying, potentially from field office closings or other administrative changes.[2] Other explanations relate to economic and demographic changes in the 2010s: population aging, which could have increased exits from SSDI through retirement and death; lower labor force participation among men, which could have made them less likely to meet the work history requirements for SSDI; a shift to less physically demanding jobs, which could have improved population health; economic recovery following the Great Recession, which could have increased work opportunities for people with disabilities; and more access to other benefit programs, which could have served as substitutes for SSDI. There are also factors that would be expected to work in the opposite direction of the decline, including the increase in the full retirement age, which would have mechanically kept SSDI recipients on SSDI longer; increases in earnings inequality, which could have increased demand for support programs; and population aging and the opioid crisis, which could have increased disability incidence.[3]

In this paper, we use statistical and causal analysis to explain the historical increase in SSDI enrollment in the 1990s and 2000s and the subsequent decline in SSDI enrollment since 2013. In Section 2, we develop a statistical decomposition method to quantify the role of various factors to the trend in SSDI enrollment. In contrast to existing methods, such as Oaxaca (1973) and Blinder (1973), our method allows us to take into account interactions across the many components of SSDI enrollment—population composition, work-history

[1]See also Glenn and Goss (2018), Social Security Administration (2019), Ruffing (2023), Center on Budget and Policy Priorities (2025), Liu and Quinby (2023), Mohamed, Burns and Cubanski (2024).

[2]See Glenn and Goss (2018), Ruffing (2023), Center on Budget and Policy Priorities (2025), vs Liu and Quinby (2023).

[3]Population aging: Ruffing (2023), Center on Budget and Policy Priorities (2025), Liu and Quinby (2023), Mohamed et al. (2024). Lower male LFP: Mohamed et al. (2024) vs Liu and Quinby (2023). Industry shift: Social Security Administration (2019) vs Liu and Quinby (2023). Economic recovery: Glenn and Goss (2018), Ruffing (2023), Center on Budget and Policy Priorities (2025), Liu and Quinby (2023). Other benefits: Glenn and Goss (2018), Ruffing (2023), Center on Budget and Policy Priorities (2025), vs Social Security Administration (2019). ALJs: Glenn and Goss (2018), Social Security Administration (2019), Ruffing (2023), Center on Budget and Policy Priorities (2025), Liu and Quinby (2023), Mohamed et al. (2024). Mailings: Ruffing (2023), Center on Budget and Policy Priorities (2025), Armour (2018), Mohamed et al. (2024). FRA: Social Security Administration (2019). Inequality: Social Security Administration (2019). Population health: Liu and Quinby (2023). Opioid crisis: Maestas, Sherry and Strand (2024).

eligibility, application, award, and various exit channels. We use our method to calculate upper and lower bounds on the contribution of each component.

In Section 3, we apply the decomposition method to individual-level administrative data from the Social Security Administration (SSA). The administrative data allow us to do a number of things that would not be possible in publicly available aggregated data: account for composition changes across a rich set of population and applicant characteristics, determine work-history eligibility for SSDI at the individual level, link SSDI applications to awards, track exit through multiple channels, and study joint outcomes (e.g., eligibility, application, and award) in addition to marginal ones.

We first apply the statistical decomposition to the increase in SSDI enrollment in the 1990s and 2000s. We find, consistent with Liebman (2015) and other analyses, that a large majority of the increase is attributable to a combination of higher entry onto SSDI and lower exit from SSDI within demographic groups, rather than changes in demographic characteristics of the population. Our data and method allow us to expand on previous studies of the increase in a number of ways. Contrary to conventional wisdom, we find that low-skilled women, rather than men, drove the increase in entry in the 1990s and 2000s.[4]

Next, we apply the statistical decomposition to the decline in SSDI enrollment in the 2010s. We find that virtually all of the decline can be attributed to declines in entry into SSDI within demographic groups, rather than increases in exit or changes in demographic characteristics of the population. Most of this decline in SSDI entry is attributable to lower application rates, with some contribution from lower award rates; changes in work-history eligibility contribute little to the decline in entry. The decline in award rates is consistent with previously documented policy changes that reduced award rates at the appeals levels (Hoynes et al., 2024), while the decline in application rates is not connected to an obvious policy change. We find that—in contrast to the increase in SSDI in the 1990s and 2000s—most of the decline in SSDI enrollment in the 2010s comes from men rather than women.

Since applications play the largest role in the decline in SSDI enrollment, we turn to a closer analysis of the decline in applications. In Section 4, we characterize the marginal non-applications during the decline period. The decline in applications comes disproportionately from applicants with higher-severity conditions and applicants who were working just prior to application. Consistent with Deshpande and Lockwood (2022), we find that higher-severity applicants are more likely to be connected to the labor market. These more-severe, more-labor-force-attached applicants become less likely to apply after the Great Recession.

[4]See, e.g., Scott Winship, "Labor-Force Participation Among Men Is Declining As Disability Claims Rise," Jan 13, 2015; Joint Economic Committee, "The Reward of Work, Incentives, and Upward Mobility," December 1, 2016.

Marginal non-applications are spread across all disability types, with musculoskeletal conditions underrepresented and mental and endocrine conditions overrepresented.

Turning to application and work outcomes, we find that low-to-middle-skilled men—who drive the decline in SSDI applications—had record-high earnings and employment following the Great Recession. Indeed, we find that during the 2010s these men became less likely to apply for SSDI and more likely to work, and that these changes are concentrated in men with health conditions. This means that any explanation for the decline in application rates must be consistent with higher labor force participation in the same population.

In Sections 5 and 6, we move beyond the accounting analysis of the decomposition to estimate the causal effects of several potential causes of the decline in SSDI applications. We start with a model of application behavior that we use to guide and interpret our analysis. We then use both existing and novel evidence to determine the causal effect of various factors on SSDI application rates. We combine the causal estimates with the model to quantify the empirical importance of the various factors for the decline in SSDI applications since 2010.

We consider how much of the decline in SSDI application rates each of the hypotheses that has been proposed can explain (conclusions in parentheses):

1. **Lower award rates at the appeals level (little, if any).** Hoynes et al. (2024) show that changes at the appeals level reduced award rates in the 2010s. We consider whether lower award rates discouraged applications, using a difference-in-differences analysis that exploits variation across counties in the decline in appeals award rates. We estimate an elasticity of applications with respect to award rates close to zero.

2. **Administrative burden of applying (little, if any).** We find that Social Security field office closings explain only a small fraction of the decline, using estimates from Deshpande and Li (2019). Measures like wait times and processing times indicate that administrative burden actually fell during this time period.

3. **Changes in population health (potentially some).** We find evidence at odds with pollution and smoking reductions as major contributors. Improvements in mental health treatment coincided with the start of the application decline and could explain some of the decline. However, the decline is not limited to mental health conditions.

4. **Opioid crisis (little, if any).** The opioid crisis could have increased SSDI applications through worse health, or could have reduced applications by killing potential applicants. However, we find no geographic heterogeneity in the decline in SSDI applications with respect to the severity of the opioid crisis.

5. **Generosity of other programs (little, if any).** There were no major expansions of SNAP, TANF, or UI during this time period. Estimates from other studies rule out health insurance as a major contributor, and we find no relationship across states between health insurance expansions and SSDI application declines. We also consider the large recent expansion of Veterans Disability Compensation, but find that veterans contribute less than proportionally to the decline in SSDI applications.

6. **Population aging (opposite direction).** Population aging during this time period actually worked against the SSDI decline, since the working-age population shifted to the highest-SSDI-receipt ages (just below the full retirement age). Only in future years will the population have aged enough to contribute to a decline in SSDI enrollment.

7. **Better employment opportunities (potentially large).** We consider whether improved employment opportunities could have reduced SSDI applications. Our analysis builds on Autor and Duggan (2003), who find that SSDI application rates are sensitive to changes in demand for less-skilled workers, especially after the 1984 expansion of medical eligibility. We replicate their research design to examine whether improvements in the labor market conditions of less-skilled men from 2010 to 2019 could have reduced SSDI application rates. Although the confidence intervals are wide, the point estimates suggest that improved employment opportunities for less-skilled men explain nearly all of the decline in SSDI application rates (relative to the counterfactual) for men, and 57 percent of the decline in application rates across all groups. To gain precision, we consider alternative specifications that use additional years of data. Even the lower end of confidence intervals of the alternative estimates suggests that changes in labor demand are a key driver of the decline in SSDI applications from 2010 to 2019.

The combination of descriptive and causal evidence in Sections 3 to 6 is consistent with better employment opportunities and inconsistent with other explanations. Improved employment opportunities in recent years appear to have pulled potential SSDI applicants into the labor market and diverted them from SSDI. A natural question is whether this phenomenon was also at work during economic recoveries in the 1990s and 2000s. As we discuss, the recovery from the Great Recession has been far longer and deeper than previous recoveries. Previous economic recoveries also coincided with policies that expanded SSDI eligibility, generosity, and program knowledge, making it difficult to isolate the effects of those recoveries.

Related literature. We make several contributions to the literature on disability benefits. We conduct the first comprehensive statistical and causal analysis of both the increase in SSDI enrollment in the 1990s and 2000s and the decline in SSDI enrollment in the 2010s.

This is also the first paper of which we are aware to use individual-level SSA administrative data in a decomposition of changes in SSDI benefits. Our paper builds on evaluations of the increase in SSDI enrollment in the 1990s and 2000s (Autor and Duggan, 2003, 2006; Reno, 2011; Congressional Budget Office, 2012; Liebman, 2015), particularly Liebman (2015), who conducts a sequential decomposition of the SSDI increase in the 1990s and 2000s using aggregated administrative data. Maestas, Mullen and Strand (2015) and Maestas et al. (2021) analyze the effects of the Great Recession on the Social Security disability program.

Our statistical decomposition method combined with individual-level administrative data on applications and awards allows for additional insights into the trends in SSDI enrollment. We build on analyses of the recent SSDI decrease (Glenn and Goss, 2018; Social Security Administration, 2019; Liu and Quinby, 2023; Ruffing, 2023; Mohamed et al., 2024; Center on Budget and Policy Priorities, 2025), particularly Liu and Quinby (2023), who use aggregated SSA data to decompose the decline into four components—population aging, the Great Recession, field office closings, and a residual term. Hoynes et al. (2024) show how changes in the appeals process in the late 2000s reduced award rates.

Our finding of improved labor market opportunities among individuals with health conditions is consistent with evidence on higher employment rates among people with disabilities in recent years. Ne'eman and Maestas (2023b) find that, after decreasing in the Great Recession, employment rates among people with disabilities began recovering in 2014. Bloom, Dahl and Rooth (2026) find that the work-from-home revolution created by the COVID-19 pandemic can potentially explain much of the rapid rise in post-pandemic employment among people with disabilities documented by Ne'eman and Maestas (2023b).

Our paper also builds on and contributes to a literature on decomposition methods in economics. These decomposition methods are best understood as an accounting approach, in which observed outcomes for one group are used to construct counterfactual scenarios for another group (Katz and Autor, 1999; Fortin, Lemieux and Firpo, 2011). In constructing these scenarios, we follow the literature on decomposition methods in abstracting from potentially important partial equilibrium considerations (e.g., selection on unobservables) and general equilibrium considerations (e.g., simultaneous determination of labor supply and the return to work).[5] Our decomposition method is tailored to sequential decision problems where actions taken at one stage (e.g., application stage) influence the options and outcomes at future stages (e.g., award stage). To the extent that application and award outcomes comove, a statistical decomposition will not know to which stage to attribute the change. The bounds produced by our decomposition method take such interaction effects into account.

[5]See Fortin et al. (2011) for a review of decomposition methods in economics and a discussion of these considerations.

1 Data and institutional background

1.1 Administrative and survey data

The data for the statistical decomposition and parts of the causal analysis come from the Social Security Administration's Continuous Work History Sample, which is a 10% sample of Social Security Numbers. The source of the SSNs is the Numident, which includes all individuals with an SSN or ITIN. The Numident thus also provides population counts for our analysis. Using the CWHS, we link individuals to the following data:

- Demographic characteristics from the Numident, including age, sex, race, and immigrant status;
- Annual earnings (1951–2022) from the Master Earnings File (MEF) Summary Earnings Record (SER), which allows us to construct work-history eligibility for SSDI in each year; average annual earnings between the ages of 25 and 34, which we use as a measure of socioeconomic status; as well as average monthly indexed earnings (AIME), primary insurance amount (PIA), and SSDI replacement rate (PIA divided by a measure of potential earnings);
- Annual earnings by employer (1978–2022) from the Master Earnings File (MEF) Detailed Earnings Record (DER), which allows us to link individuals to firm information such as industry from the Department of Labor F5500 using Employer Identification Number (EIN);
- SSDI applications from the F831 file (1990–2022), which include application date, initial (DDS) decision date, initial (DDS) decision, the "step" at which the applicant was allowed or denied, body system code, primary and secondary diagnosis, medical diary reason (how often the approved applicant is supposed to get a medical review in the future, which we use as a measure of severity), years of education, state, and ZIP code (for some years);
- Additional application information from MEDIB (2005-2019), including ZIP code and legal representation;
- OASDI payments from the Payment History Update System (PHUS) at the monthly level (1984–2022), which serve as our primary measure of SSDI enrollment; and
- SSDI enrollment data from the Master Beneficiary Record (MBR) (1970–2022), which allow us to measure exit from SSDI via recovery (i.e., medical review).

Using this administrative data, we do several things that would not be possible in publicly available aggregated SSDI data: calculate work-history eligibility for each individual in the sample, link SSDI applications to awards to calculate application rates and award rates by year of application, track individuals from SSDI entry to SSDI exit, and document the type of exit (death, retirement, recovery, or other). Most importantly, the SSA administrative data allow us to study joint outcomes (e.g., eligibility, application, and award for the same individual), rather than just marginal outcomes.

For individuals with multiple applications, we assign the award to the last application before the award. For cases in which there is an award but no application (a small fraction of cases), we assign the application year to be the award year.

We measure enrollment rate and award rate based on the year of the application to which they are linked (i.e., the award rate is the fraction of applications in a given year that are ultimately allowed). Our enrollment rate and award rate numbers differ slightly from publicly reported numbers because the public numbers do not link applications to awards; instead, they divide the number of awards in a year by the number of (unrelated) applications in that year. Appendix Figure D.10 compares our application-linked award rates to the non-linked award rates in the public data. Appendix Figure G.1 compares our measure of enrollment counts to the actual CWHS data.

Tracking applications submitted in a given year requires administrative data for several years after the application year, since many SSDI applications take years to go through the decision and appeals processes. Our last year of administrative data is 2024, which means we can reliably track applications submitted as late as 2021. Therefore our decomposition exercise can only be applied to years 2021 and earlier.

In Appendix Section D.1, we show that our data match public data on enrollment, eligibility, application, award, and exit numbers well. Appendix Figures D.9 and D.10 demonstrate the difference between calculating award rates using applications and award made in the same year, versus linking applications in a given year to whether they are ever allowed in the future.

We also use a 1% version of the CWHS that includes ZIP codes from the Master Earnings File in order to calculate changes in primary insurance amount across geographic areas.

We link information from publicly available data sources to get additional characteristics of individuals and geographic areas:

- Department of Labor Form 5500: We link DOL Form 5500 to the DER using EIN to get industry (NAICS) codes for the industry in which an individual worked early in life and immediately before applying. We use this measure to study heterogeneity by early-career industry and recent industry.

- Publicly available military EINs: Military EINs are publicly available.[6] We use the military EINs to identify veterans. We use this measure to study heterogeneity by veteran status.

- Pollution: We use data provided by Isen, Rossin-Slater and Walker (2017) to identify individuals born in places that were either required or not required to abate pollution under the Clean Air Act of 1970. We use this measure to study heterogeneity by local pollution abatement.

- Remote work: We use data on the prevalence of working from home by industry (NAICS codes) constructed by Dingel and Neiman (2020).[7] We link individuals to this measure of remote work prevalence in the industry they worked in just before applying for SSDI using NAICS codes from the F5500 (see above). We use this measure to study heterogeneity by the availability of remote work.

- ZIP code and county level data, linked to applicants in the F831:

  - Opioid deaths by county (1999–2020) from the Centers for Disease Control and Prevention Wonder tool;

  - Social mobility measures by county from Chetty, Hendren, Kline, Saez and Turner (2014b);[8]

  - Unemployment rate by county from 2010–2017 from the Department of Labor Bureau of Labor Statistics; and

  - Minimum wage data by state and locality collected by Vaghul and Zipperer (2022).[9]

We complement the administrative data file with Survey of Income and Program Participation (SIPP) linked to SSA administrative data. Although the sample size is much smaller, the SIPP has many useful features for our analysis. Most importantly, it includes measures of health status for the full population that are lacking in administrative data. It also provides information on spouses and family members and measures of educational attainment. Appendix Section B.1 provides more details on the SIPP analysis.

[6] See, e.g., `https://www.dfas.mil/Portals/98/Documents/Garnishments/06_Quick_Guide_201603.pdf?ver=2020-02-20-133132-973`.

[7] See `https://github.com/jdingel/DingelNeiman-workathome`.

[8] See `https://opportunityinsights.org/paper/land-of-opportunity/`

[9] See `https://github.com/benzipperer/historicalminwage/releases/tag/v1.4.0`.

1.2 Background on the SSDI program

Social Security Disability Insurance (SSDI) is intended to provide assistance to Americans who cannot work because of a severe health condition. An individual is considered disabled if (i) "[they] cannot do work that [they] did before," (ii) "[SSA] decide[s] that [they] cannot adjust to other work because of [their] medical condition(s)," and (iii) "[their] disability has lasted or is expected to last for at least one year or to result in death."[10] Non-health factors such as age and education may also be considered in the SSDI decision through the medical-vocational grids.

SSDI eligibility is limited to individuals with qualifying earnings histories. To qualify for SSDI, individuals between the ages of 30 and 42 years old must have at least 20 quarters of qualifying earnings. The required number of quarters increases by one for every year past age 42. Workers under 30 qualify with fewer credits.

SSDI benefits are increasing in past earnings, at a decreasing rate. The average annual SSDI cash benefit is around $15,000, and beneficiaries qualify for Medicare after a two-year waiting period.[11] Disabled workers are not permitted to earn more than the substantial gainful activity (SGA) level, which was $1,260 per month ($15,120 per year) in 2020.

Supplemental Security Income (SSI) is another disability program administered by the Social Security Administration. SSI has the same medical requirements as SSDI, but different non-medical requirements: SSI requires low income and assets, while SSDI requires a work history. SSI enrollment rates have also been declining since 2013.

2 Statistical decomposition method

Our goal is to decompose the difference between actual SSDI enrollment and counterfactual SSDI enrollment into the share attributable to various components: changes in composition across groups, and changes in behavior (including eligibility, application, award, and exit) within groups. To do the decomposition, we first define counterfactual SSDI enrollment by deciding which parameters to hold fixed at their baseline values, and which parameters to allow to evolve over time. We construct counterfactual SSDI enrollment by simulating what SSDI enrollment would have been in future years had certain parameters (e.g., SSDI entry and exit rates) been held fixed at their baseline values. We then decompose the difference between raw and counterfactual SSDI enrollment into various components.

[10] See `https://www.ssa.gov/benefits/disability/qualify.html`.

[11] Annual Statistical Report on the Social Security Disability Insurance Program, 2018.

2.1 Defining counterfactual SSDI enrollment

We start by defining counterfactual SSDI enrollment. We can write SSDI enrollment as an accounting equation, where enrollment among working-age adult SSDI beneficiaries at the end of year t is determined by enrollment among SSDI beneficiaries in year $t-1$, net of entry and exit that occurred in year t. Letting $S_{t,g}$ denote SSDI enrollment ("stock") among those in calendar year t with fixed observable characteristics g ("group"), this accounting identity can be expressed as:

$$S_{t,g} = S_{t-1,g} + N_{t,g}(P_{t-1,g} - S_{t-1,g}) - X_{t,g}S_{t-1,g} \tag{1}$$

where $P_{t,g}$ denotes the working-age population count with characteristics g in year t, and $N_{t,g}$ and $X_{t,g}$ denote their rate of entry into and exit from SSDI, respectively. Exit includes death, aging into the retirement program, recovery (i.e., determined by medical review), and all other reasons. In our baseline exercises, we define g to represent individuals with a common birth cohort, sex, socioeconomic status, and immigrant status.[12]

From a start year normalized as $t = 0$, we characterize the SSDI enrollment that would have arisen in future year t given some counterfactual path of entry and exit flows between years 0 and t by recursive substitution of equation (1):

$$S_{t,g}(\mathbf{N}_g, \mathbf{X}_g, S_{0,g}) = \sum_{i=0}^{t} N_{t-i,g}P_{t-i,g}K(g,i,t) + S_{0,g}K(g,t+1,t) \tag{2}$$

$$K(g,i,t) \equiv \begin{cases} \prod_{j=1}^{i}(1 - N_{t-j+1,g} - X_{t-j+1,g}) & \text{if } i \geq 1 \\ 1 & \text{if } i = 0 \end{cases}$$

where $\mathbf{N}_g$ and $\mathbf{X}_g$ denote vectors of entry and exit flows, each of length $t+1$, and $K(g,i,t)$ captures how existing SSDI stocks transmit from year i to year t. The counterfactual SSDI enrollment can be translated into a counterfactual SSDI enrollment rate ($O_{t,g}$) given the population count in year t: $O_{t,g}(\mathbf{N}_g, \mathbf{X}_g, S_{0,g}, P_g) \equiv \frac{S_{t,g}(\mathbf{N}_g, \mathbf{X}_g, S_{0,g})}{P_g}$. The aggregate counterfactual SSDI enrollment rate can be expressed as a population-weighted average of the group-specific rates:

$$O_t(\mathbb{N}, \mathbb{X}, \mathbf{S}_0, \mathbf{P}) = \sum_g O_{t,g}(\mathbf{N}_g, \mathbf{X}_g, S_{0,g}, P_g) \times w_{t,g}, \quad w_g \equiv \frac{P_g}{\sum_g P_g} \tag{3}$$

As seen in equations (2) and (3), aggregate SSDI enrollment rates are a non-linear func-

[12] We use average annual earnings between ages 25 and 34 years as a proxy for socioeconomic status. We divide individuals into 5 bins: \$0, above \$0 but less than \$10,000, above \$10,000 but less than \$30,000, above \$30,000 but less than \$60,000, and above \$60,000.

tion of entry flows, exit flows, and the distribution of observables across all intervening years. Still, it is useful to decompose them into additively separable components that can be attributed to each component.

To do that, we compare the actual path of SSDI enrollment to a counterfactual benchmark in which entry rates and exit rates (conditional on sex, socioeconomic status, and *age*) are held fixed at their initial, year-0 values. This is the counterfactual SSDI enrollment we would have expected in year t had no aging, sex composition changes, or changes in the income distribution occurred, and if SSDI entry rates and exit rates conditional on observables had remained stable over time. To be precise, we define:

$$N^c_{s,g} = N^r_{0,a(g,s)}, \quad X^c_{s,g} = X^r_{0,a(g,s)}, \quad w^c_{s,g} = w^r_{0,a(g,s)} \quad \forall \quad 0 \leq s \leq t, \quad g \tag{4}$$

where c denotes the counterfactual value and r denotes the actual value. $a(g, s)$ denotes the group sharing the same socioeconomic status and sex as group g, but born s years earlier than group g; thus, group $a(g, s)$ is the same age in year 0 as group g is in year s. We plug these values into equation (2) to construct the counterfactual path of SSDI enrollment.[13]

2.2 Decomposing SSDI enrollment into components of interest

Having constructed the counterfactual path of SSDI enrollment, we decompose the difference in actual SSDI enrollment and counterfactual SSDI enrollment into various components of interest.

Changes in SSDI due to changes in observables: Oaxaca-Blinder. The conventional Oaxaca-Blinder decomposition allows us to attribute the difference between actual and counterfactual SSDI enrollment rates into changes in SSDI enrollment conditional on population characteristics, changes in the composition of population characteristics, and their interaction:

[13]Population counts $P_{t,g}$ should follow their own accounting equation, depending on mortality and net migration of SSDI beneficiaries and non-beneficiaries. In deriving these expressions, we assume $P^c_{t,g} \approx P^r_{t,g}$ for simplicity and omit the superscript. In practice though, we account for how changing the mortality and net migration of SSDI beneficiaries can affect population counts, and track it using an additional term in the decomposition: a "population friction" accounting for the fact that O^r_t and O^c_t have different denominators. This component is small because mortality rates for SSDI beneficiaries are generally low, and SSDI beneficiaries are a small fraction of the overall population.

$$O_t^r - O_t^c = \underbrace{\sum_g O_{t,g}^c (w_{t,g}^r - w_{t,g}^c)}_{\text{change in characteristics}} + \underbrace{\sum_g (O_{t,g}^r - O_{t,g}^c) w_{t,g}^c}_{\text{change from conditional SSDI rates}} + \underbrace{\sum_g (O_{t,g}^r - O_{t,g}^c)(w_{t,g}^r - w_{t,g}^c)}_{\text{characteristics} \times \text{SSDI rate interaction}} \tag{5}$$

Note that the terms that we attribute to characteristics and to conditional SSDI rates are *symmetric*; each holds its counterpart components fixed at their counterfactual values (c as opposed to r). The interaction term reflects the ambiguity inherent in how SSDI rates can be attributed to these two factors when both are changing at the same time. We can load this interaction term into the first two terms in order to arrive at alternative attributions. For instance, if we load the interaction term fully into the second term (attributing it to changes in group SSDI rates), then we arrive at an *asymmetric*, two-term Oaxaca-Blinder decomposition where population weights are fixed at their true observed values (r) when judging the impact of changes in conditional SSDI rates.

Changes in SSDI due to changes in entry and exit flows. Equation (5) allows us to assess how aggregate SSDI rates have been driven by changes in SSDI rates fixing observable characteristics, but not how those rates have been driven by changes in entry flows versus exit flows. To accomplish that, consider the year-1 difference in counterfactual versus real SSDI rates for each group g. Using equation (1), we can express that difference in terms of additively separable components attributable to changes in entry rates versus changes in exit rates:

$$O_{1,g}^r - O_{1,g}^c = \frac{S_{1,g}^r - S_{1,g}^c}{P_{1,g}} = \underbrace{(N_{1,g}^r - N_{1,g}^c)\frac{(P_{0,g} - S_{0,g})}{P_{1,g}}}_{\text{change from conditional entry rate}} - \underbrace{(X_{1,g}^r - X_{1,g}^c)\frac{S_{0,g}}{P_{1,g}}}_{\text{change from conditional exit rate}}$$

where we use that $S_{0,g}^c = S_{0,g}^r = S_{0,g}$ by construction.

Recursion for $t > 0$. For years of the exercise beyond the first, decomposing the counterfactual difference in SSDI rates becomes complicated by the fact that pre-existing differences in SSDI stocks accumulated in prior years can contribute to determining the impact of *current* differences in entry and exit flows. Using the stock accounting equation (equation 1) and rearranging terms, we can write for any $t > 0$:

$$O^r_{t+1,g} - O^c_{t+1,g} = \frac{S^r_{t+1,g} - S^c_{t+1,g}}{P^c_{t+1,g}} + S^r_{t+1,g}\left(\frac{1}{P^r_{t+1,g}} - \frac{1}{P^c_{t+1,g}}\right) \approx \frac{S^r_{t+1,g} - S^c_{t+1,g}}{P_{t+1,g}} \tag{6}$$

$$\begin{aligned} S^r_{t+1,g} - S^c_{t+1,g} \approx & \underbrace{(N^r_{t,g} - N^c_{t,g})(P_{t,g} - S^c_{t,g})}_{a_{t,g}\text{: change in conditional entry}} \quad \underbrace{-(X^r_{t,g} - X^c_{t,g})S^c_{t,g}}_{b_{t,g}\text{: change in conditional exit}} \\ & + \underbrace{(1 - N^c_{t,g} - X^c_{t,g})(S^r_{t,g} - S^c_{t,g})}_{c_{t,g}\text{: change in past conditional SSDI rate}} \\ & + \underbrace{(N^c_{t,g} - N^r_{t,g})(S^r_{t,g} - S^c_{t,g})}_{d_{t,g}\text{: entry} \times \text{past SSDI rate interaction}} \quad \underbrace{-(X^r_{t,g} - X^c_{t,g})(S^r_{t,g} - S^c_{t,g})}_{e_{t,g}\text{: exit} \times \text{past SSDI rate interaction}} \end{aligned}$$

This expression involves new interaction terms ($d_{t,g}$, $e_{t,g}$) which describe how the impact of counterfactual changes in entry and exit depends on initial SSDI stocks (or equivalently, how counterfactual changes in SSDI stocks carry over from year to year in a way that depends on entry and exit rates). The stated approximations come from observing that $P^c_{t,g} \approx P^r_{t,g}$ in practice.[14] In this section, we use this approximation to focus our decomposition only on its key components. See Appendix Section E for the exact version of the decomposition which we implement in practice.

The natural next step is to recursively substitute for past differences in SSDI rates ($S^r_{t,g} - S^c_{t,g}$) to arrive at an expression that depends only on counterfactual differences in entry and exit rates. However, each recursive step introduces new interaction terms, reflecting that past differences SSDI stocks at some point arose due to a difference in entry rates or exit rates. It is therefore useful for tractability to take a stand on where to assign those interactions. If we attribute the past-rate interaction terms ($d_{t,g}$) and ($e_{t,g}$) to changes in past SSDI rates ($c_{t,g}$), we arrive at a simpler expression:

$$\begin{aligned} S^r_{t+1,g} - S^c_{t+1,g} \approx & \underbrace{\sum_{i=0}^{t} (N^r_{t-i,g} - N^c_{t-i,g})(P_{t-i,g} - S^c_{t-i,g})K^r_{g,i,t}}_{\text{cumulative change from conditional entry rate}} \\ & \underbrace{- \sum_{i=0}^{t} (X^r_{t-i,g} - X^c_{t-i,g})S^c_{t-i,g}K^r_{g,i,t}}_{\text{cumulative change from conditional exit rate}} \end{aligned} \tag{7}$$

[14] As stated in Appendix E, the only difference between $P^c_{t,g}$ and $P^r_{t,g}$ comes from differences in the mortality rates of SSDI beneficiaries. This has a very limited effect on population counts because mortality rates are always fairly low and SSDI beneficiaries are a small fraction of the overall population.

where $K_{g,i,t}$ is defined as in equation (2), rearranging its arguments as subscripts.

Notice that we can formulate other versions of this recursive decomposition by loading the interaction terms instead into conditional entry rates and exit rates.[15] The choice of how to allocate the interaction terms forms the basis of our bounding exercise, outlined in detail in Appendix Section F. To construct our bounds, we restrict to decompositions in which these interaction terms are assigned in a way that is consistent over time and across groups. That is, if we assign term $d_{t,g}$ to changes in conditional entry for one particular t and g, then we do the same *for all* t and g. Our bounds therefore nest many ways to decompose SSDI stocks into additively separable terms, but are not guaranteed to nest all reasonable linear decompositions of SSDI rates without conditions on how flows have changed over time.[16] We specify the full range of estimates that we construct for this step and the remaining steps to follow in Appendix Section F.

Further decomposing changes in exit and entry. Distinguishing the source of changes in exit rates is straightforward as total exits are simply a sum of exits due to the three observed exit channels (retirement, mortality, and recovery) plus all other exits.

$$X_{t,g} = \text{Retirement Rate}_{t,g} + \text{Mortality Rate}_{t,g} + \text{Recovery Rate}_{t,g} + \text{Other Exit Rate}_{t,g} \quad (8)$$

Decomposing entry is more complicated. We can decompose entry into SSDI further into contributions from work-history eligibility, applications conditional on eligibility, awards conditional on applying, and the interaction between three terms. Recall that the definition of $N_{t,g}$ is the entry rate of non-SSDI beneficiaries who existed in the year-initial of year t for group g. We can write:

$$\begin{aligned} N_{t,g} &= \frac{\text{entrants}_{t,g}}{\text{initial non-DI beneficiaries}_{t,g}} \\ &= \underbrace{\frac{\text{entrants}_{t,g}}{\text{applicants}_{t,g}}}_{B_{t,,g}=\text{award}_{t,g}} \underbrace{\frac{\text{applicants}_{t,g}}{\text{elig pop}_{t,g}}}_{A_{t,g}=\text{application}_{t,g}} \underbrace{\frac{\text{elig pop}_{t,g}}{P_{t-1,g} - S_{t-1,g}}}_{E_{t,g}=\text{eligibility}_{t,g}} \end{aligned}$$

[15] We continue to assume here that $P^c_{t,g} \approx P^r_{t,g}$, meaning that changing SSDI exits due to mortality does not meaningfully affect the population. In practice, that difference creates an additional term in the decomposition that we label "population difference," which captures the idea that the impact of entry rates in year $t-i$ on SSDI stocks in year t depends on the size of the surviving population in year $t-i$.

[16] One notable decomposition approach which is not nested in our set of decompositions is a "sequential" decomposition as in Liebman (2015). However, each term of a sequential decomposition is guaranteed to lie in the range of values generated by our decomposition for that term, so long as changes in entry and exit flows are both driving an increases in SSDI rates or decreases in SSDI rates for all t. This is approximately true for the periods of time we examine.

We decompose the difference in entry rates $(N^r_{t,g} - N^c_{t,g})$ into differences in award rates $B_{t,g}$, application rates $A_{t,g}$, and eligibility rates $E_{t,g}$:

$$\begin{aligned}
N^r_{t,g} - N^c_{t,g} =& B^r_{t,g}A^r_{t,g}E^r_{t,g} - B^c_{t,g}A^c_{t,g}E^c_{t,g} \\
=& \underbrace{(B^r_{t,g} - B^c_{t,g})A^c_{t,g}E^c_{t,g}}_{\text{partial award}} + \underbrace{B^c_{t,g}(A^r_{t,g} - A^c_{t,g})E^c_{t,g}}_{\text{partial apply}} + \underbrace{B^c_{t,g}A^c_{t,g}(E^r_{t,g} - E^c_{t,g})}_{\text{partial elig}} \\
&+ \underbrace{B^c_{t,g}(A^r_{t,g} - A^c_{t,g})(E^r_{t,g} - E^c_{t,g})}_{\text{apply} \times \text{elig}} \\
&+ \underbrace{(B^r_{t,g} - B^c_{t,g})A^c_{t,g}(E^r_{t,g} - E^c_{t,g})}_{\text{award} \times \text{elig}} \\
&+ \underbrace{(B^r_{t,g} - B^c_{t,g})(A^r_{t,g} - A^c_{t,g})E^c_{t,g}}_{\text{award} \times \text{apply}} \\
&+ \underbrace{(B^r_{t,g} - B^c_{t,g})(A^r_{t,g} - A^c_{t,g})(E^r_{t,g} - E^c_{t,g})}_{\text{award} \times \text{apply} \times \text{elig}}
\end{aligned} \tag{9}$$

where "partial" means the part that can be attributed only to that component and no other. Substituting this expression for $(N^r_{t-i,g} - N^c_{t-i,g})$ in equation (7), we arrive at a decomposition of SSDI stocks that distinguishes between the impact of eligibility, applications among the eligible, and awards among applicants. Substituting the contribution of change in stock into equation (6) and subsequently equation (5), the contribution of type g to the application component of the enrollment rate change is (analogous equations for eligibility, award, entry, and exit in Appendix Section E):

$$\frac{w^c_{t,g}}{P_{t,g}} \sum_{i=0}^{t} B^c_{t-i,g}(A^r_{t-i,g} - A^c_{t-i,g})E^c_{t-i,g}(P_{t-i,g} - S^c_{t-i,g})K^r_{g,i,t} \tag{10}$$

Decomposition of changes in characteristics. We consider how to attribute the change due to observable characteristics as a whole into the change due to a specific observable characteristic—e.g., sex, aging, or socioeconomic status. The population weight for group g amounts to the joint density evaluated at g, which in our main exercise represents the share of the population in a specific age (a), sex (s), and socioeconomic bin (e) in year t. We can express this weight as a product of conditional probabilities:

$$w_{t,g} = w_{t,s\times a\times e} = \underbrace{\sum_{a,e} w_{t,s\times a\times e}}_{P(s|t)\equiv\Omega_s} \underbrace{\frac{\sum_{e} w_{t,s\times a\times e}}{\sum_{a,e} w_{t,s\times a\times e}}}_{P(a|s,t)\equiv b_{s,a}} \underbrace{\frac{w_{t,s\times a\times e}}{\sum_{e} w_{t,s\times a\times e}}}_{P(e|a,s,t)\equiv f_{s,a,e}}$$

We can then decompose counterfactual differences in these weights into additively separable terms reflecting the impact of changes in the marginal sex distribution, the conditional distribution of age given sex, the conditional distribution of socioeconomic status given sex and age, and interaction terms:

$$\begin{aligned} w^r_{t,g} - w^c_{t,g} &= w^r_{t,s\times a\times e} - w^c_{t,s\times a\times e} \\ &= (\Omega^r b^r f^r) - (\Omega^c b^c f^c) \\ \text{(sex composition)} &= (\Omega^r - \Omega^c) b^c f^c \\ \text{(age composition fixing sex)} &+ \Omega^c (b^r - b^c) f^c \\ \text{(SES fixing age and sex)} &+ \Omega^c b^c (f^r - f^c) \\ &+ \text{ (interaction terms)} \end{aligned} \tag{11}$$

The interaction terms consist of four components: three representing the possible ways that any two covariates could jointly change, and one representing joint changes in all three covariates. However, note that the order of covariates in this sequence is important for the magnitude and interpretation of the terms, as conditional and unconditional changes in the composition can be very different. For instance, the unconditional impact of changes in sex composition (as in the ordering of equation (11)) is empirically close to zero because the sex distribution of the working-age population has not evolved much over time. If we instead swap the positions of sex and socioeconomic status in the decomposition, then we would find a modest contribution from changes in the composition of sex *conditional on age and socioeconomic status* as men are increasingly likely to have relatively high earnings early in their working careers.

2.3 Partitioning application and award changes into subgroups

It may also be informative to understand which subgroups are driving changes in application rates, award rates, and overall entry rates. To do so, we partition the sample of applications into mutually exclusive subgroups (e.g., low-skilled men) given by g.

We partition the change in unconditional application rate ($\tilde{A}^r_t - \tilde{A}^c_t$) into subgroups:

$$\begin{aligned}
\tilde{A}_t^r - \tilde{A}_t^c &= \sum_g A_{g,t}^r E_{g,t}^r w_{g,t}^r - A_{g,t}^c E_{g,t}^c w_{g,t}^c \\
&= \sum_g \underbrace{A_{g,t}^c E_{g,t}^c (w_{g,t}^r - w_{g,t}^c)}_{\text{Across group composition}} + \sum_g \underbrace{(A_{g,t}^r E_{g,t}^r - A_{g,t}^c E_{g,t}^c)(w_{g,t}^r - w_{g,t}^c)}_{\text{Within-across interaction}} + \sum_g (A_{g,t}^r E_{g,t}^r - A_{g,t}^c E_{g,t}^c) w_{g,t}^c \\
&= \sum_g \underbrace{A_{g,t}^c E_{g,t}^c (w_{g,t}^r - w_{g,t}^c)}_{\text{Across group composition}} + \sum_g \underbrace{(A_{g,t}^r E_{g,t}^r - A_{g,t}^c E_{g,t}^c)(w_{g,t}^r - w_{g,t}^c)}_{\text{Within-across interaction}} \qquad (12) \\
&+ \sum_g \underbrace{(A_{g,t}^r - A_{g,t}^c) E_{g,t}^c w_{g,t}^c}_{\text{Application rate contribution}} + \sum_g \underbrace{A_{g,t}^c (E_{g,t}^r - E_{g,t}^c) w_{g,t}^c}_{\text{Eligibility rate contribution}} + \sum_g \underbrace{(A_{g,t}^r - A_{g,t}^c)(E_{g,t}^r - E_{g,t}^c) w_{g,t}^c}_{\text{Interaction within apply}}
\end{aligned}$$

We partition the change in unconditional entry rate ($\tilde{N}_t^r - \tilde{N}_t^c$) into subgroups:

$$\begin{aligned}
\tilde{N}_t^r - \tilde{N}_t^c &= \sum_g \underbrace{N_{g,t}^r}_{\text{Cond. entry rate}} w_{g,t}^r - N_{g,t}^c w_{g,t}^c \\
&= \sum_g \underbrace{N_{g,t}^c (w_{g,t}^r - w_{g,t}^c)}_{\text{Across group composition}} + \sum_g \underbrace{(N_{g,t}^r - N_{g,t}^c)(w_{g,t}^r - w_{g,t}^c)}_{\text{Within-across interaction}} + \sum_g \underbrace{(N_{g,t}^r - N_{g,t}^c) w_{g,t}^c}_{\text{entry rate contribution}} \\
\sum_g (N_{g,t}^r - N_{g,t}^c) w_{g,t}^c &= \sum_g (B_{g,t}^r A_{g,t}^r E_{g,t}^r - B_{g,t}^c A_{g,t}^c E_{g,t}^c) w_{g,t}^c \\
&= \sum_g \underbrace{(B_{g,t}^r - B_{g,t}^c) A_{g,t}^c E_{g,t}^c w_{g,t}^c}_{\text{Award rate contribution}} + \sum_g \underbrace{B_{g,t}^c (A_{g,t}^r - A_{g,t}^c) E_{g,t}^c w_{g,t}^c}_{\text{Application rate contribution}} \qquad (13) \\
&+ \sum_g \underbrace{B_{g,t}^c A_{g,t}^c (E_{g,t}^r - E_{g,t}^c) w_{g,t}^c}_{\text{Eligibility rate contribution}} + \underbrace{\cdots}_{\text{Interaction within entry}}
\end{aligned}$$

We partition the change in award rate conditional on application ($\tilde{B}_t^r - \tilde{B}_t^c$) into subgroups as follows, where $F_{g,t} \equiv \dfrac{\text{applicants}_{g,t}}{\sum_g \text{applicants}_{g,t}}$ is the fraction of total applicants that subgroup g makes up:

$$\begin{aligned}
\tilde{B}_t^r - \tilde{B}_t^c &= \sum_g B_{g,t}^r F_{g,t}^r - B_{g,t}^c F_{g,t}^c \\
&= \sum_g \underbrace{(B_{g,t}^r - B_{g,t}^c) F_{g,t}^c}_{\text{Award rate contribution}} + \sum_g \underbrace{(F_{g,t}^r - F_{g,t}^c) B_{g,t}^c}_{\text{Applicant fraction contribution}} + \sum_g (B_{g,t}^r - B_{g,t}^c)(F_{g,t}^r - F_{g,t}^c)
\end{aligned} \qquad (14)$$

3 Decomposition results

3.1 Applying decomposition method to SSDI increase (1988–2010)

We start by applying our decomposition method to the increase in SSDI enrollment from 1988–2010. Table 1 presents the main results, which are largely consistent with Liebman (2015) (see Appendix Table A.1 for a comparison). Higher within-group entry rates account for most (55-58%) of the increase in the overall SSDI enrollment rate, but lower within-group exit rates (23-25%) are also an important part of the story. Changes in composition also play a nontrivial role (18-23%) in the SSDI enrollment rate increase. Table 1 shows that the bounds on the main components are relatively tight. Appendix Figure A.1 shows the decomposition results over time.

Table 1: Decomposition results for the increase in SSDI enrollment from 1988 to 2010

	Absolute contribution (pp)			As a share of total (%)		
	Minimum	Maximum	Used in graph	Minimum	Maximum	Used in graph
Across-group composition	**0.39**	**0.49**	**0.39**	**18.0**	**22.5**	**18.0**
Sex	-0.11	0.03	0.00	-5.3	1.3	0.1
Age	0.41	1.08	0.47	18.8	49.7	21.5
Earnings	-0.30	0.18	0.09	-13.7	8.3	4.3
Immigrant status	-0.32	-0.13	-0.19	-14.6	-5.9	-8.8
Within-group SSDI rates	**1.69**	**1.79**	**1.69**	**77.5**	**82.0**	**77.5**
Entry	1.20	1.26	1.20	54.8	57.6	54.8
Eligibility	*0.15*	*0.21*	*0.15*	*7.1*	*9.5*	*7.1*
Application	*0.64*	*0.80*	*0.80*	*29.5*	*36.6*	*36.6*
Award	*0.17*	*0.28*	*0.17*	*8.0*	*12.9*	*8.0*
Exit	0.51	0.55	0.51	23.3	25.0	23.3
Death	*0.32*	*0.38*	*0.32*	*14.9*	*17.6*	*14.9*
Recovery	*0.06*	*0.06*	*0.06*	*2.5*	*2.7*	*2.5*
Retirement	*0.12*	*0.13*	*0.13*	*5.4*	*6.0*	*6.0*
Other	*-0.01*	*-0.00*	*-0.00*	*-0.6*	*-0.2*	*-0.2*
Friction in population	-0.02	-0.02	-0.02	-1.0	-1.0	-1.0
Population discrepancies	0.01	0.01	0.01	0.4	0.4	0.4
Within-across interaction			**0.10**			**4.5**
Total to be explained	2.18	2.18	2.18	100.0	100.0	100.0

Notes: Table presents decomposition of the increase in SSDI enrollment rates from 1988 to 2010 into various components. The counterfactual SSDI enrollment rate in 2010 is 2.43 percent while the actual enrollment rate in 2010 is 4.61 percent, creating 2.18 pp of SSDI increase to be decomposed (see Appendix Table G.1). Across-group composition change, within-group change in SSDI enrollment rate, and the interaction between the two are calculated using equation (5). Across-group composition change is decomposed into specific characteristics using equation (11). Within-group SSDI rate is decomposed into entry, exit, friction in population, and population discrepancies using equation (6) and footnotes 13 and 15. Entry is decomposed into partial eligibility (i.e., the part that can only be attributed to eligibility and no other component), partial application, partial award, and the sum of interactions between the three entry components (eligibility, application, and award) using equation (9). Exit is decomposed into death, retirement (defined as aging into the retirement program), recovery (defined as being found no longer eligible due to continuing disability review), and "other" (the residual) using equation (8). "Minimum" means the minimum bound, "Maximum" means the maximum bound, and "Used in graph" means the point estimate used in Appendix Figure A.1. See Section 2.1 for details on the decomposition method, and Appendix Section F for details on the construction of bounds.

The administrative panel data allow us to expand on the findings in Liebman (2015). In Figure 2, using equation (13), we find that the entry rate increase from 1988 to 2010 came disproportionately from low-skilled women (both younger and older), while men and higher-skilled individuals played a smaller role. The finding that women drove the vast majority of the entry from 1988 to 2010 contrasts with the conventional narrative about low-skilled men

driving SSDI enrollment in the 1990s and 2000s.[17]

Figure 2: Increase in SSDI entry (1988–2010) driven by low-skilled women



(a) Entry rate increase

(b) Entrants in 1988

Notes: Left graph plots the fraction of the contribution of changes (in absolute terms) in entry rate that can be attributed to specific demographic groups, as given by equation (13). Negative numbers indicate that the group worked *against* the increase. The point estimates we report are the ones in which flows (e.g., B_t, A_t, E_t, X_t), when fixed, are held fixed at their counterfactual values. Right graph plots the composition of entrants in 1988 as a baseline comparison. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old. See Appendix Table G.8 for exact estimates.

3.2 Applying decomposition method to SSDI decline (2010–2019)

We now apply our decomposition method to the decrease in SSDI enrollment in the 2010s. In the main analysis, we start the exercise in 2010, since applications began declining in that year; and end the exercise in 2019, in order to avoid the unusual circumstances of the COVID-19 pandemic (when, e.g., Social Security field offices closed for two years). However, we probe the robustness of the results to using alternative start years (e.g., 2013, when

[17]See, e.g., Scott Winship, "Labor-Force Participation Among Men Is Declining As Disability Claims Rise," Jan 13, 2015; Joint Economic Committee, "The Reward of Work, Incentives, and Upward Mobility," December 1, 2016. However, as shown in Appendix Table A.1, the decomposition analysis in Liebman (2015) also finds that women accounted for more of the increase in SSDI enrollment than men.

enrollment began declining) and end years (e.g., 2021, post-pandemic). It is important to note that the "end" year in the analysis requires several more years of data beyond that year, since we track applications filed in that year their to final outcome (e.g., initial decision, appeal, final decision). Our last year of administrative data is 2024, which means we can reliably track applications submitted as late as 2021.

Table 2 and Figure 3 present the main results. We find that virtually all of the decline in SSDI enrollment is attributable to decreases in within-group entry rates (90-97%), rather than changes in composition (1-5%) or changes in exit (1-4%). Within entry, most of the decrease comes from lower application rates (46-57%) and lower award rates (30-43%). The role of award rates is perhaps expected given that the decline in award rates at the appeals level has been well-documented and attributed in part to retraining of administrative law judges (Hoynes et al., 2024). To our knowledge, however, there is no policy change that obviously explains the decline in applications. Moreover, as we discuss in more detail in Section 4.1, the decomposition is likely to underestimate the contribution from application rate changes (and overestimate the contribution from award rate changes) due to a shift in applicant composition. Once we adjust for changing applicant composition by severity, the share attributable to application increases from 46-57% to 57-69%; the share attributable to award falls by the equivalent amount (see Appendix Table A.4).

Which demographic groups drive the decline in application rates? In Figure 4, using equation (12), we find that men, especially younger men, contribute disproportionately to the decline in SSDI application rates. Older women contribute the least to the application decline, in both absolute terms and relative to their baseline shares. Looking by industry, the application rate decline comes from all industries, though construction and manufacturing are the largest contributors relative to their baseline shares (see Appendix Section G.4).

We do a similar exercise for award rates (conditional on applying) in Figure 5. The decline in award rates is driven disproportionately by younger women; this pattern is likely explained by the ALJ retraining reducing award rates most for mental conditions, which younger women are more likely to apply with. Older men contribute the least to the award rate decline. The decline in entry rates (a combination of application rate and award rate) is driven disproportionately by younger people, both men and women (Appendix Table G.22).

Robustness of decomposition to alternative start and end years. Our main decomposition exercise spans 2010 to 2019: we choose 2010 as the start year since it is the year in which applications start declining, and 2019 as the end year since it is the last year before the COVID-19 pandemic created extenuating circumstances (e.g., the two-year-long closure of all SSA field offices).

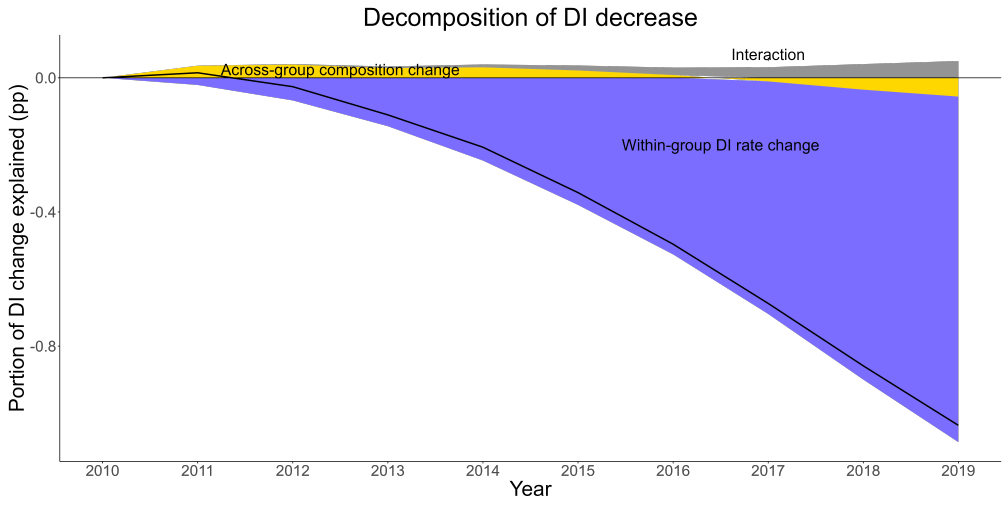
Table 2: Decomposition results for the decrease in SSDI enrollment from 2010 to 2019

	Absolute contribution (pp)			As a share of total (%)		
	Minimum	Maximum	Used in graph	Minimum	Maximum	Used in graph
Across-group composition	**-0.01**	**-0.06**	**-0.06**	**0.6**	**5.4**	**5.4**
Sex	0.00	-0.02	0.00	-0.3	2.1	-0.0
Age	0.18	0.02	0.07	-17.6	-1.9	-6.7
Earnings	0.03	-0.11	-0.04	-3.1	10.4	3.5
Immigrant status	-0.06	-0.17	-0.11	5.9	15.9	10.6
Within-group SSDI rates	**-0.98**	**-1.03**	**-1.03**	**94.6**	**99.4**	**99.4**
Entry	-0.94	-1.00	-0.99	90.3	96.9	95.2
Eligibility	*-0.05*	*-0.09*	*-0.08*	*4.5*	*8.4*	*8.1*
Application	*-0.48*	*-0.59*	*-0.58*	*46.1*	*56.8*	*55.6*
Award	*-0.31*	*-0.45*	*-0.44*	*29.9*	*43.0*	*42.0*
Exit	-0.01	-0.04	-0.04	1.3	3.9	3.5
Death	*0.04*	*0.04*	*0.04*	*-4.2*	*-3.5*	*-4.1*
Recovery	*-0.06*	*-0.08*	*-0.08*	*5.6*	*7.4*	*7.3*
Retirement	*0.01*	*0.01*	*0.01*	*-1.0*	*-0.7*	*-0.8*
Other	*-0.01*	*-0.01*	*-0.01*	*0.6*	*1.1*	*1.1*
Friction in population	-0.01	-0.01	-0.01	0.9	1.1	0.9
Population discrepancies	0.00	0.00	0.00	-0.2	-0.1	-0.1
Within-across interaction			**0.05**			**-4.8**
Total to be explained	-1.04	-1.04	-1.04	100.0	100.0	100.0

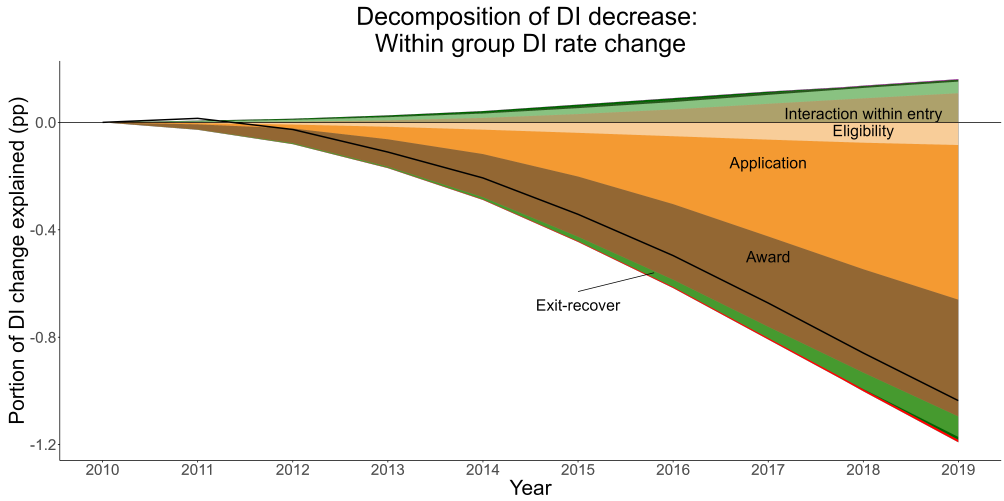
Notes: Table presents decomposition of the decrease in SSDI enrollment rates from 2010 to 2019 into various components. The counterfactual SSDI enrollment rate in 2019 is 5.37 percent while the actual enrollment rate in 2019 is 4.33 percent, creating 1.04 pp of SSDI decline to be decomposed (see Appendix Table G.9). Across-group composition change, within-group change in SSDI enrollment rate, and the interaction between the two are calculated using equation (5). Across-group composition change is decomposed into specific characteristics using equation (11). Within-group SSDI rate is decomposed into entry, exit, friction in population, and population discrepancies using equation (6) and footnotes 13 and 15. Entry is decomposed into partial eligibility (i.e., the part that can only be attributed to eligibility and no other component), partial application, partial award, and the sum of interactions between the three entry components (eligibility, application, and award) using equation (9). Exit is decomposed into death, retirement (defined as aging into the retirement program), recovery (defined as being found no longer eligible due to continuing disability review), and "other" (the residual) using equation (8). "Minimum" means the minimum bound, "Maximum" means the maximum bound, and "Used in graph" means the point estimate used in Figure 3. See Section 2.1 for details on the decomposition method, and Appendix Section F for details on the construction of bounds.

However, we probe the robustness of the decomposition results to alternative start and end years. First, we change the start year to 2013, since SSDI enrollment began declining in 2013 (versus 2010 for applications). Using 2013 as the start year, we find that even more of the SSDI decline is attributed to application (58-62%) and less to award (13-17%) (Appendix Table A.2). Second, we change the end year to 2021 to see how much the COVID-19 pandemic mattered. Using 2021 as the end year produces results nearly identical to the main results (Appendix Table A.3). Note that an "end" year of 2021 means we track applications filed in 2021 or earlier to their final outcome as of 2024, since some applications take several years to process.

Figure 3: SSDI decrease (2010–2019) attributable to lower application and award



(a) Across-group versus within-group changes



(b) Within-group changes in behavior

Notes: Top figure plots the decomposition of the decrease in SSDI enrollment rates from 2010 to 2019 into across-group composition change, within-group change in SSDI enrollment rate, and the interaction between the two, as given in equation (5). Bottom figure plots the decomposition of the within-group change in SSDI enrollment rate into specific components of SSDI entry, SSDI exit, friction in the population, and population discrepancy. The specific components of SSDI entry are partial eligibility (i.e., the part that can only be attributed to eligibility and no other component), partial application, partial award, and the sum of interactions between the three entry components (eligibility, application, and award), as given in equation (9). The specific components of SSDI exit are partial death, partial retirement (defined as aging into the retirement program), partial recovery (defined as being found no longer eligible due to continuing disability review), and partial other (any remaining exit after removing death, retirement, and recovery), as given in equation (8). "Friction in population" and "Population discrepancy" are described in footnotes 13 and 15. See Section 2.1 for details on the decomposition method. The point estimates we report are the ones in which flows (e.g., B_t, A_t, E_t, X_t), when fixed, are held fixed at their counterfactual values. See Table 2 for exact estimates.

Figure 4: Application rate decline from 2010 to 2019 driven by men



(a) Application rate decrease

(b) Applicants in 2010

Notes: Left graph plots the fraction of application rate decline that can be attributed to specific demographic groups, per equation (12). The point estimates we report are the ones in which flows (e.g., B_t, A_t, E_t, X_t), when fixed, are held fixed at their counterfactual values. Right graph plots the composition of applicants in 2010 as a baseline comparison. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old. See Appendix Table G.16 for exact estimates.

Figure 5: Award rate decline from 2010 to 2019 driven disproportionately by women



(a) Award rate decrease

(b) Entrants in 2010

Notes: Left graph plots the fraction of award rate decline that can be attributed to specific demographic groups, per equation (14). The point estimates we report are the ones in which flows (e.g., B_t, A_t, E_t, X_t), when fixed, are held fixed at their counterfactual values. Right graph plots the composition of entrants in 2010 as a baseline comparison. The demographic groups are characterized by sex, average annual earnings between ages 25-34 (a measure of skill), and whether older or younger than 45 years old. See Appendix Table G.19 for exact estimates.

4 Characterizing the marginal (non-)applications and (non-)awards during the SSDI decline

In the previous section, we find that the largest share of the decline is attributable to a decline in within-group SSDI application rates, driven disproportionately by men and especially younger men. Most of the remainder of the SSDI enrollment decline is attributable to a decline in within-group award rates, driven disproportionately by younger women. In this section, we characterize the marginal (non-)applications and (non-)awards over this time period using additional characteristics.

4.1 Decomposing application and award rates using characteristics observed only for applicants

The SSA application data include detailed characteristics on applicants that are not observed for the full population, including severity, diagnosis, and geographic location. Since we do not observe these characteristics for the full population, we cannot incorporate them into the decomposition method outlined above. However, under certain assumptions, we can use them to characterize the applications (and awards) that would have taken place in the counterfactual world but did not in the real world.

We partition applications into mutually exclusive subgroups based on an applicant-specific characteristic d (e.g., for diagnosis, partition into mental, musculoskeletal, and other conditions). We write the difference between the counterfactual application rate for (population-level) group g at time t ($A^c_{t,g}$) and the real application rate ($A^r_{t,g}$) in terms of the d subgroups:

$$A^r_{t,g} - A^c_{t,g} = \sum_d \hat{A}^r_{t,g,d} - \sum_d \hat{A}^c_{t,g,d} = \sum_d (\hat{A}^r_{t,g,d} - \hat{A}^c_{t,g,d}) \tag{15}$$

where $\hat{A} \equiv \frac{applicant_{d,g,t}}{eligpop_{g,t}} \equiv P(A = 1, d | t, g, E = 1)$ is the joint conditional probability of applying and belonging to subgroup d, given year-group (t, g) and conditional on SSDI eligibility ($E = 1$). Changes in $\hat{A}_{t,g,d}$ could be driven by changes in the probability of applying conditional on belonging to subgroup d (e.g., having a musculoskeletal condition), or changes in the prevalence of subgroup d in the SSDI-eligible population (e.g., in the prevalence of musculoskeletal conditions for year-group t, g). If we assume that the subgroup composition in the SSDI-eligible population is stable over time, then changes we attribute to subgroup d using equation (15) are driven by changes in application rates among individuals

in subgroup d (e.g., changes in application rates conditional on having a musculoskeletal condition). These assumptions appear to be plausible during the period of the SSDI decline: we find no changes in eligibility rates conditional on observables (see Section 3), and no improvements in population health (see Appendix Figures A.11 and A.12).[18]

We can similarly decompose the award rate $B_{t,g}$ using mutually exclusive subgroups:

$$\begin{aligned} B^r_{t,g} - B^c_{t,g} = & \underbrace{\sum_d (b^r_{t,g,d} - b^c_{t,g,d}) a^c_{t,g,d}}_{\text{change in } d\text{-specific award rates}} + \underbrace{\sum_d (a^r_{t,g,d} - a^c_{t,g,d}) b^c_{t,g,d}}_{\text{change in application composition}} \\ & + \underbrace{\sum_d (b^r_{t,g,d} - b^c_{t,g,d})(a^r_{t,g,d} - a^c_{t,g,d})}_{\text{interaction term}} \end{aligned} \tag{16}$$

since $B_{t,g} = \sum_d b_{d,g,t} a_{d,g,t}$ where $b_{t,g,d}$ is the subgroup-specific award rate (e.g., award rates for applications coming from mental, musculoskeletal, or other conditions) and $a_{t,g,d}$ is the share of applications belonging to subgroup d (e.g., the share of applications from year-group (t, g) coming from mental, musculoskeletal, or other conditions).

The "change in application composition" term could be driven by two different channels: changes in characteristics in the SSDI-eligible population, or changes in application behaviors conditional on subgroup. Interpreting the composition term in equation (16) as coming purely from changes in application behaviors requires the same assumptions we needed above to interpret equation (15)—specifically, that subgroup composition has not changed in the SSDI-eligible population. Under this assumption, the composition term represents how changes in applicant composition across subgroups mechanically affects award rates. In contrast, the "change in d-specific award rates" term reflects changes in award rates *within* subgroup, perhaps attributable to policy changes.

To decompose entry by applicant-specific characteristics, we start by writing the difference in real and counterfactual entry rates as follows using the definitions of $a_{d,g,t}$ and $b_{d,g,t}$ from above:

[18] Interpreting changes in application characteristics as changes in selection into application also requires the assumption that there have been no changes in how SSA measures the applicant characteristics. Some characteristics like geography are objective. For more subjective characteristics like diagnosis and severity, there have been no policy changes during the period of SSDI decline.

$$
\begin{aligned}
N^r_{g,t} - N^c_{g,t} &= B^r_{g,t}A^r_{g,t}E^r_{g,t} - B^c_{g,t}A^c_{g,t}E^c_{g,t} \\
&= \sum_d b^r_{d,g,t}a^r_{d,g,t}A^r_{g,t}E^r_{g,t} - b^c_{d,g,t}a^c_{d,g,t}A^c_{g,t}E^c_{g,t} \\
&= \sum_d (b^r_{d,g,t} - b^c_{d,g,t})a^c_{d,g,t}A^c_{g,t}E^c_{g,t} + \sum_d b^c_{d,g,t}(a^r_{d,g,t} - a^c_{d,g,t})A^c_{g,t}E^c_{g,t} \\
&+ \sum_d b^c_{d,g,t}a^c_{d,g,t}(A^r_{d,g,t} - A^c_{g,t})E^c_{g,t} + \sum_d b^c_{d,g,t}a^c_{d,g,t}A^c_{g,t}(E^r_{d,g,t} - E^c_{g,t}) + ...
\end{aligned} \tag{17}
$$

If we plug this into the "entry rate contribution" part of equation (13) and sum over groups g, we can write:

$$
\begin{aligned}
\sum_g (N^r_{g,t} - N^c_{g,t})w^c_{g,t} &= \sum_g \underbrace{(B^r_{g,t} - B^c_{g,t})A^c_{g,t}E^c_{g,t}w^c_{g,t}}_{\text{Award rate contribution}} + \sum_g \underbrace{B^c_{g,t}(A^r_{g,t} - A^c_{g,t})E^c_{g,t}w^c_{g,t}}_{\text{Application rate contribution}} \\
&+ \sum_g \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})w^c_{g,t}}_{\text{Eligibility rate contribution}} + \underbrace{...}_{\text{Interaction within entry}} \\
&= \sum_{gd} \underbrace{(b^r_{dgt} - b^c_{dgt})a^c_{dgt}A^c_{gt}E^c_{gt}w^c_{g,t}}_{\text{Award (by } d\text{)}} + \sum_{gd} \underbrace{(a^r_{gt} - a^c_{dgt})b^c_{dgt}A^c_{gt}E^c_{gt}w^c_{g,t}}_{\text{Applicant share (by } d\text{)}} \\
&+ \sum_{gd} \underbrace{(b^r_{dgt} - b^c_{dgt})(a^r_{dgt} - a^c_{dgt})A^c_{gt}E^c_{gt}w^c_{g,t}}_{\text{Interaction within award and apply (by d)}} \\
&+ \sum_{gd} \underbrace{b^c_{dgt}a^c_{dgt}(A^r_{gt} - A^c_{gt})E^c_{gt}w^c_{g,t}}_{\text{Application (by } d\text{)}} + \sum_g \underbrace{B^c_{g,t}A^c_{g,t}(E^r_{g,t} - E^c_{g,t})w^c_{g,t}}_{\text{Eligibility}} \\
&\underbrace{...}_{\text{Interaction terms}}
\end{aligned} \tag{18}
$$

where the second equality uses equation (16) to decompose the "award rate contribution" term.

4.2 Characteristics of marginal non-applications

Using equation (15), we analyze heterogeneity in the decline in SSDI applications with respect to several different applicant characteristics. Figure 6 shows results with respect to three key characteristics for the decline from 2010 to 2019: severity, pre-application employment, and diagnosis. For each characteristic and time period, the top bar represents the actual breakdown of the change in application rate. The bottom bar represents what the breakdown *would have been hypothetically* if it had been proportional to the shares of

applicant characteristics in 2007.

Perhaps the most striking finding is that with respect to applicant severity. The decline in applications during this period comes disproportionately from the moderately-severe and most-severe categories, as opposed to the least-severe. Marginal non-applicants are also more likely to be recently attached to the labor force. While the application decline comes from all disability types, applications for musculoskeletal conditions are underrepresented while applications with mental conditions and endocrine conditions (e.g., diabetes) are overrepresented.

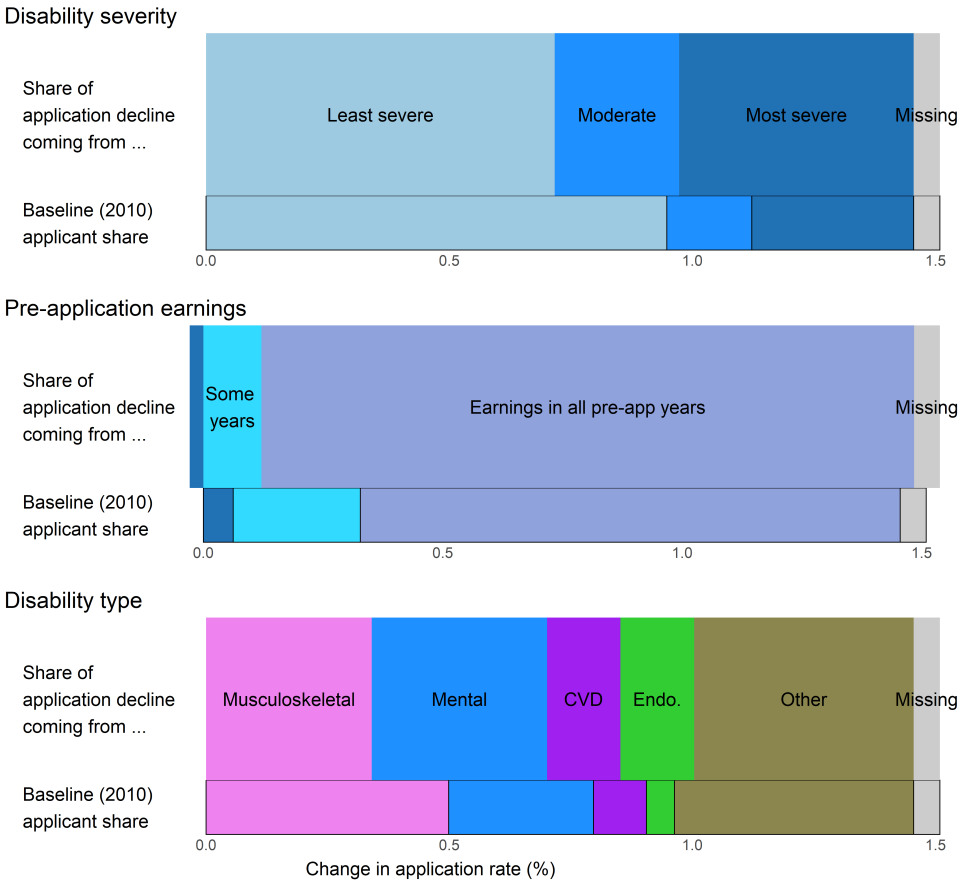
Diving deeper into these results, we find that applicant severity is positively correlated with recent applicant earnings: in 2019, the "most severe" applicants were more likely (71%) to have recently worked full-time than the "least severe" applicants (42%), and were more likely (38% vs 21%) to have early-career earnings in the top two earnings quintiles (see Appendix Figure A.2). The "least severe" appear more likely to apply primarily for economic reasons rather than health reasons, and are highly likely to be rejected. The most common diagnoses for the "most severe" are musculoskeletal (35%, mostly disorders of the back, arthropathies, and arthritis), cancer (15%), and neurological (13%); for the "least severe," they are musculoskeletal (39%), mental (22%), and neurological (8%). These facts suggest that middle- and high-severity applicants are the marginal workers, while low-severity applicants are much more disconnected from the labor market.[19]

We conduct heterogeneity analysis with respect to several other characteristics, with results in Appendix Section H.2. Marginal non-applicants are slightly more likely to come from areas with improving local labor market conditions (Appendix Table H.16). They are disproportionately from the construction and manufacturing industries (Appendix Table H.25), and less likely to be in industries with high work-from-home rates (Appendix Table H.26). However, we find no meaningful geographic heterogeneity with respect to state, urban/rural status, minimum wage changes, severity of the opioid crisis, economic mobility as measured by Chetty, Hendren, Kline and Saez (2014a), or the decline in SSDI allowance rates at the appeals level (Appendix Tables H.12, H.14, H.15, H.19, H.20, and H.21). The composition of the decline in entry (i.e., approved applications) looks similar to the composition of the decline in applications, but less disproportionate for nearly all characteristics.

Finally, we compare the marginal applications during the Great Recession (2007-2014) period to the marginal non-applications during the post-recovery period (2014-2019). As shown in Appendix Figure A.3, we find that these two groups of marginal applicants are

[19]This finding is consistent with Deshpande and Li (2019) and Deshpande and Lockwood (2022). Our primary measure of severity uses the medical diary reason, which is the disability examiner's assessment of how often the condition should be reviewed for continuing medical eligibility. We also use the "step" at which an individual is allowed or denied as an alternative measure of severity (see Appendix Table H.11).

Figure 6: Marginal non-applicants disproportionately severe, labor-force-attached, and non-musculoskeletal



Notes: Figure characterizes the change in applicants by characteristics during 2010-2019. Each bar consists of a main bar and a sub-bar. The main bar gives the fraction of the decline from 2010-2019 represented by that characteristic, using equation (15). The sub-bar gives the hypothetical fractions that would have been proportional to applicant shares in 2010. "Least severe" means that the applicant was rejected at the disability examiner level (but could have been allowed on appeal). "Moderately severe" means that the disability examiner assigned a diary reason of "medical improvement expected" or "medical improvement possible." "Most severe" means that the disability examiner assigned a diary reason of "medical improvement not expected." "Earnings in all pre-app years" means the applicant worked for 4-5 years of the 5 years before application. "Earnings in some pre-app years" means the applicant worked for 1-3 years of the 5 years before application. "No earnings" means the applicant did not work in any of the 5 years before application. Diagnosis is recorded by the disability examiner assigned to the case. See Appendix Tables H.3, H.6, and H.27 for exact estimates.

quite different. During the Great Recession period, the marginal applications are disproportionately from areas with declining labor market conditions, disproportionately in the least-severe category, and disproportionately representing musculoskeletal and mental conditions. These results are consistent with Maestas et al. (2015) and Maestas et al. (2021). In contrast, during the post-recession period, the marginal non-applications are disproportionately from areas with strong economic growth and in higher-severity categories, but are

relatively proportional in the types of conditions they apply with. The Great Recession period appears to have pulled in applicants with less-severe conditions and musculoskeletal and mental conditions onto SSDI, while the post-recession period appears to have discouraged applicants with more-severe conditions and non-musculoskeletal conditions.

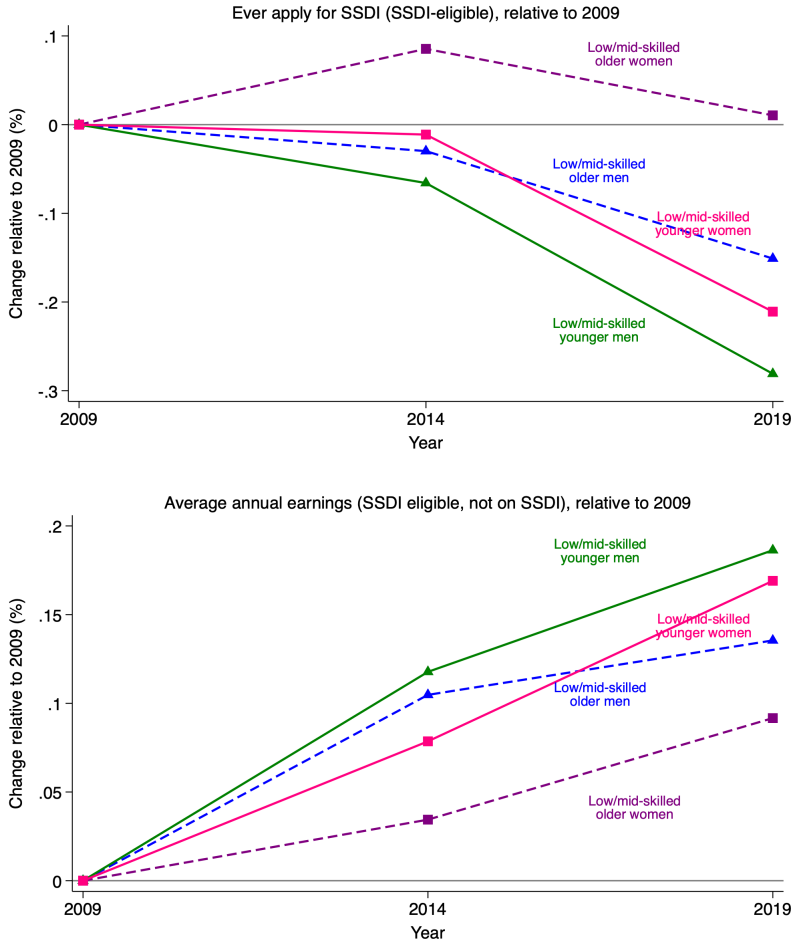
Adjusting the decomposition results for changes in applicant composition. The change in applicant composition with respect to severity during the 2010-2019 decline means that the statistical decomposition is likely to underestimate the share of the decline in SSDI enrollment attributable to declining application rates. Since the applicant pool becomes less severe over time, award rates will mechanically fall. The statistical decomposition will attribute the entire decline in award rates causally to changes in award rates, when in fact some of the decline in award rates is the result of changing applicant composition. To account for differential selection into application over time, we further decompose the "Award rate contribution" term in equation (18) into the parts attributable to true changes in award rates versus changes arising from changing applicant composition. This adjustment increases the share of the SSDI enrollment decline attributable to lower application rates from 46-57% to 57-69%, and decreases the share attributable to lower award rates from 30-43% to 19-31% (see Appendix Table A.4).

4.3 Characterizing the work behavior of marginal non-applicants

Figure 7 shows SSDI application rates and inflation-adjusted earnings for different (low-to-middle-skilled) demographic groups during the application decline period. Consistent with the results in Figure 4, younger men experience the largest declines in application rates, followed by younger women and older men, while older women experience no decline in application rates. The patterns in earnings gains almost perfectly mirror the application results: younger men experience the largest gains, followed by younger women and older men, with older women lagging behind. In Appendix Figure A.4, we find similar results by early-career industry: workers who started out in construction and manufacturing have the largest application declines and the largest earnings gains over this time.

How do we reconcile large earnings gains among younger men with previous findings on declining male labor force participation? In Appendix Figure A.5, we find that the SSA administrative data replicate the finding from previous work that non-employment rates have increased substantially for younger low-to-middle-skilled men. However, since SSDI requires a substantial work history, non-employed men are not on the margin for SSDI receipt. The large earnings gains in Figure 7 are driven by younger men who are attached to the labor

Figure 7: Low/mid-skilled younger men have largest SSDI application declines and largest earnings gains



Notes: Top figure plots the likelihood of ever applying for SSDI, relative to 2009. Bottom figure plots average annual earnings relative to 2009. "Low/mid-skilled" means average annual earnings of $1-$30,000 between the ages 25 to 34 years. "Younger/older" means age 35-39 and 50-54, respectively. The sample is derived from the Continuous Work History Sample linked to the Numident, F831 file, Payment History Update System, and Master Earnings File.

market and therefore eligible for SSDI.[20]

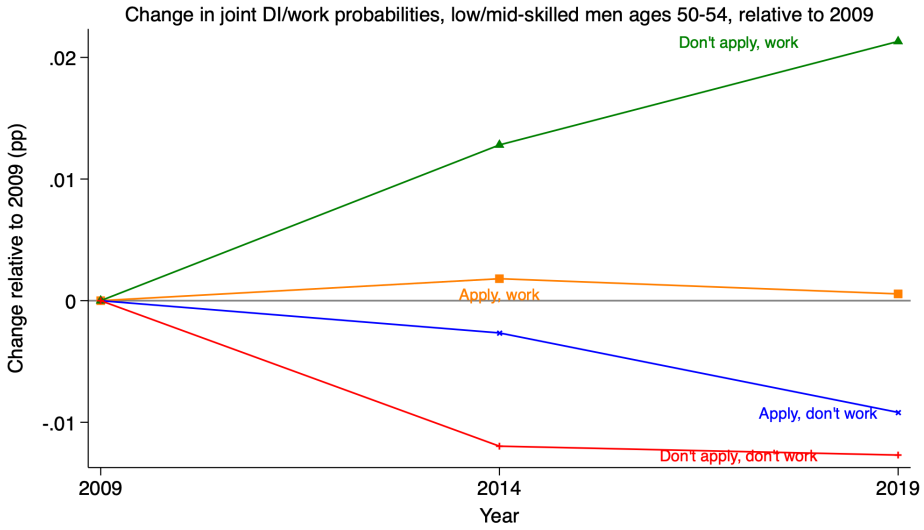
A key question is whether the workers with higher earnings are the *same* workers who are less likely to apply for SSDI. We conduct two exercises to shed light on this question. First, we test whether the increase in earnings and employment among low-skilled men applies to those with health conditions, who are most at risk for SSDI receipt. Since the administrative data do not have health measures for the full population, we use SIPP-SSA

[20]Other sources also document historic earnings growth at the bottom of the wage distribution following the Great Recession (Shambaugh and Strain, 2021; Aeppli and Wilmers, 2022; Dey, Handwerker, Piccone Jr and Voorheis, 2022). More generally, the post-Great-Recession recovery has been the longest and strongest recovery in recent decades (see Appendix Figure A.6).

linked data to condition our analysis on health status reported in the SIPP. In Appendix Figure A.7, we present results from the same exercise as in Figure 7, but for individuals with health conditions in the SIPP. We find even stronger patterns: among low-to-mid-skilled individuals with health conditions, younger men experience the largest application declines, followed by older men and younger women, and then by older women; the earnings gains mirror these patterns. The SIPP-SSA linked data also allow us to observe spouses, and we find that spouse earnings for low-to-middle-skilled men have grown rapidly since the Great Recession (Appendix Figure A.7). These results indicate that the increase in earnings and employment (and household resources) applies to low-skilled men who are at risk of applying for SSDI. Appendix Section B.1 provides more details on the SIPP analysis.

Second, we calculate the joint likelihood of employment and SSDI application across cohorts of low-to-middle-skilled men. Figure 8 shows that the likelihood of applying and not working has declined for older low-to-middle-skilled men in the post-recession period. At the same time, the likelihood of working and not applying has increased, while the likelihood of neither working nor applying has declined. Appendix Figure A.8 replicates the same result for older low-to-middle-skilled men with health conditions in the SIPP-SSA linked data. These results suggest that men who would have applied for SSDI instead work.

Figure 8: Low/mid-skilled men have become less likely to apply, more likely to work



Notes: Figure plots the joint probabilities of working and applying for SSDI benefits for low-to-middle-skilled men (defined as having average annual earnings of $1-$30,000 between the ages 25 to 34 years) ages 50 to 54 over time. Figure uses men from the Continuous Work History Sample linked to the Numident, F831 file, Payment History Update System, and Master Earnings File.

4.4 Characterizing marginal non-awards

Recall from the previous section that the decline in conditional award rates is driven disproportionately by younger women. We can additionally characterize the marginal non-awards by characteristics that are only observed for applicants using equation (16), with results in Appendix Section H.2. We find that the award rate decline comes disproportionately from mental conditions (Appendix Table H.8). This result is consistent with administrative law judges reducing award rates more for less-verifiable conditions as a result of their re-training. Younger men and women are disproportionately likely to apply with mental conditions, which could explain why they disproportionately drive the award rate decline.

4.5 Implications of descriptive analysis

We find that the change in application behavior is driven disproportionately by the groups that experience the largest gains in earnings. This descriptive finding suggests some reasons for the decline in SSDI enrollment and rules out other reasons. One possibility is that improved labor market opportunities caused a reduction in SSDI applications. Another possibility is that a reduction in SSDI applications—caused by, e.g., higher administrative burden, lower award rates, or stigma—led to higher labor supply. A third possibility is that some third factor, like improvements in health, led to both higher employment rates and lower SSDI application rates.

The relationship between SSDI application and work behavior is at odds with other hypotheses for the decline in SSDI applications. For example, higher generosity of other benefits might reduce SSDI applications but likely would not increase employment. In the next section, we study in more detail hypotheses for the decline in SSDI applications.

5 Causal analysis of the decline in SSDI applications

We now evaluate the empirical relevance of several potential causes of the decline in SSDI applications. We first present a model of application behavior that we will use to guide and interpret this analysis. Next, we present a set of causal estimates on SSDI application of plausibly exogenous changes in a wide range of factors that could affect incentives to apply for SSDI. The factors we consider are summarized in Table 3. Lastly, we combine the causal estimates with the model to quantify the empirical importance of the various factors to the observed decline in SSDI applications since 2010.

5.1 Model of application behavior

The model serves three purposes. First, it clarifies the parameters that we need to recover to quantify the empirical importance of the various factors of the decline in SSDI applications. Second, it helps to make explicit the data, observable shifters, and assumptions that we need to identify and estimate these parameters. Lastly, it makes it possible to construct logically coherent predictions of counterfactual trends in SSDI applications in the absence of a particular factor. By comparing this counterfactual trend to the actual trend, we can draw inferences about the empirical importance of each factor.

Preferences. Consider a population of individuals of different types ν who vary in the characteristics that determine their capacity to work. The types are distributed across individuals according to the cumulative distribution function F_{ν_t} with support V. Let A and B be indicator variables for applying to and receiving benefits, respectively. The fixed cost of applying is given by the parameter $\kappa(\nu)$. The K-dimensional vectors z_t^b and z_t^w are shifters that affect utility when on benefits and working, respectively.

The state-specific utility function of a given individual of type ν is given by:

$$V_t(A,B) = \begin{cases} \sum_{k=1}^{K} \beta_k^b(\nu) z_{kt}^b + \xi_t^b - \kappa(\nu) & \text{if } A=1 \text{ and } B=1, \\ \sum_{k=1}^{K} \beta_k^w(\nu) z_{kt}^w + \xi_t^w - \kappa(\nu) & \text{if } A=1 \text{ and } B=0, \\ \sum_{k=1}^{K} \beta_k^w(\nu) z_{kt}^w + \xi_t^w & \text{if } A=0 \text{ and } B=0, \end{cases} \tag{19}$$

where ξ_t^b and ξ_t^w are drawn i.i.d. from a standard Gumbel distribution, capturing idiosyncratic variation across individuals in the utility of working relative to receiving benefits.

Before deriving the decision rule of the model, it is useful to note that we suppress the subscript for individuals to simplify the notation. However, z_t^b, z_t^w, ξ_t^b and ξ_t^w are allowed to vary across individuals of the same type ν, while the parameters κ, β_k^w, and β_k^b vary by ν but not between individuals with the same ν.

Decision rule and application rates. At the beginning of each period, the individual forms her (rational) expectations about the award function $G_t(v) \equiv \mathbb{P}(B_t = 1 | A_t = 1, \nu_t = v)$. The function captures the leniency of the SSDI system in the sense that a higher $G_t(v)$ means that a larger share of applicants of type v are awarded.[21] The individual then decides whether or not to apply for SSDI benefits to maximize their expected utility given the state-specific utility function in equation (19) and their beliefs.

[21] Eligibility can be built into G_t by, for example, putting $G_t(v) = 0$ for certain values of v, so only individuals with certain characteristics (e.g., age or work history) are eligible.

The solution to this maximization problem leads to the following decision rule: An individual applies whenever the expected utility gain from applying is greater than the fixed cost:

$$A_t = \mathbb{1}\left[G_t(v) \times \left(\sum_{k=1}^{K}(\beta_k^b(v)z_{kt}^b - \beta_k^w(v)z_{kt}^w) + \xi_t^b - \xi_t^w\right) \geq \kappa(v)\right].$$

The application rate of individuals of type v can then be expressed in terms of the model as:

$$p_t(v) \equiv \mathbb{P}(A_t = 1 \mid \nu_t = v) = \frac{\exp(m_t(v))}{1+\exp(m_t(v))} \equiv H(m_t(v)).$$

where $m_t(v)$ captures the leniency-adjusted payoff to applying for SSDI for type v individuals:

$$m_t(v) \equiv \sum_{k=1}^{K}(\beta_k^b(v)z_{kt}^b - \beta_k^w(v)z_{kt}^w) - \frac{\kappa(v)}{G_t(v)}.$$

A model-based decomposition of changes in application rates. We are interested in understanding the causes of the long-run decline in application rates:

$$\begin{aligned}\mathbb{P}(A_{t_1}=1) - \mathbb{P}(A_{t_0}=1) &= \sum_{l=t_0}^{t_1-1}(\mathbb{P}(A_{l+1}=1) - \mathbb{P}(A_l=1)) \\ &= \sum_{l=t_0}^{t_1-1}(\underbrace{\int (f_{\nu_{l+1}}(v) - f_{\nu_l}(v))p_l(v)dv)}_{\text{changes in composition}} + \underbrace{\int (p_{l+1}(v) - p_l(v))f_{\nu_{l+1}}(v)dv)}_{\text{changes in application behavior}}\end{aligned}$$

where f_{ν_t} is the probability density function of ν_t. The first term after the second equality tells us the contribution from changes in the distribution of ν, keeping the application behavior fixed. The second term captures the contribution from changes in application behavior, keeping the distribution of ν fixed. This term can be further decomposed into three distinct components of changes in application behavior over time:

$$\int (p_{t+1}(v) - p_t(v))f_{\nu_{t+1}}(v)dv = \sum_{k=1}^{K}\Delta_{kt}^b + \sum_{k=1}^{K}\Delta_{kt}^w + \Delta_t^G$$

which (up to a first-order approximation) are defined as follows:

$$\Delta^b_{kt} = \underbrace{\int \underbrace{(f_{\nu_{t+1}}(v)h(m_t(v))}_{\text{share of } \nu \text{ type at the margin of applying}} \times \underbrace{\beta^b_k(v)}_{\text{effect on application incentives}} \times \underbrace{(z^b_{kt+1} - z^b_{kt})}_{\text{time shift in } k\text{'th benefit shifter}})dv}_{\text{aggregate change in application behavior due to time shift in the } k\text{'th benefit shifter}}$$

$$\Delta^w_{kt} = \underbrace{\int \underbrace{(f_{\nu_{t+1}}(v)h(m_t(v))}_{\text{share of } \nu \text{ type at the margin of applying}} \times \underbrace{-\beta^w_k(v)}_{\text{effect on application incentives}} \times \underbrace{(z^w_{kt+1} - z^w_{kt})}_{\text{time shift in } k\text{'th work shifter}})dv}_{\text{aggregate change in application behavior due to time shift in the } k\text{'th work shifter}}$$

$$\Delta^G_t = \underbrace{\int \underbrace{(f_{\nu_{t+1}}(v)h(m_t(v))}_{\text{share of } \nu \text{ type at the margin of applying}} \times \underbrace{\frac{\kappa(v)}{G_t(v)^2}}_{\text{effect on application incentives}} \times \underbrace{(G_{t+1}(v) - G_t(v))}_{\text{time shift in award function}})dv}_{\text{aggregate change in application behavior due to time shift in screening stringency}}$$

and $h(m) \equiv dH/dm$.

The contribution to $\mathbb{P}(A_{t_1} = 1) - \mathbb{P}(A_{t_0} = 1)$ of each component is increasing in the number of individuals that are close to the application margin, in the effect on application incentives of an increase in the component's shifter, and in the change over time in this shifter.

Taking the model-based decomposition to the data. In Section 6, we will use the model to construct logically coherent predictions of counterfactual trends in SSDI application rates in the absence of observed time changes in the k'th benefit shifter z^b_{kt}, in the k'th work shifter z^w_{kt}, or in the award function $G_t(v)$. To construct these counterfactual trends, we need to recover the parameters of Δ^b_{kt}, Δ^w_{kt}, or Δ^G_t.

In either case, the first step is to recover $f_{\nu_t}(v)$ and $h(m_t(v))$ for each year t. For this purpose, it is useful to observe that there is a tight link between the model-based decomposition above and the statistical decomposition of the time trend in the application rates. In fact, these decompositions coincide if one assumes that ν_t can be represented by the observed characteristics we use in the statistical decomposition. Under this assumption of selection on observables, we can directly measure $f_{\nu_t}(v)$ and $p_t(v)$ from the yearly micro data on SSDI applications. The quantity $h(m_t(v))$ can then be identified from the share of applicants by

inverting this equation:

$$m_t(v) = \log\left(\frac{p_t(v)}{1-p_t(v)}\right).$$

The next step is to recover the parameter $\beta_k^b(v)$, $\beta_k^w(v)$, or $\kappa(v)$. To do so, we will use a research design that isolates plausibly exogenous variation in observable shifters. However, for notational simplicity, suppose for now that the shifter z_{kt}^b, z_{kt}^w, or Δ_t^G is randomly assigned across individuals in at least one year. We can then recover $\beta_k^w(v)$ by estimating the effect of a marginal increase in z_{kt}^w on individuals of each type ν, since

$$\underbrace{\frac{\partial \mathbb{P}(A_t = 1 \mid \nu_t = v)}{\partial z_{kt}^w}}_{\text{Identified from (quasi)-experiment}} = \underbrace{h(m_t(v))}_{\text{Identified from application data}} \times \underbrace{\beta_k^w(v)}_{\text{Unobserved}}.$$

The same logic allows us to identify $\beta_b^b(v)$ by estimating the effect of a marginal increase in z_{kt}^b and recovering $h(m_t(v))$ from application data. By comparison, to recover $\kappa(v)$, we need to estimate the effect of a marginal increase in G_t on individuals of each type ν, since

$$\underbrace{\frac{\partial \mathbb{P}(A_t = 1 \mid \nu_t = v)}{\partial G_t(v)}}_{\text{Identified from (quasi)-experiment}} = \underbrace{h(m_t(v))}_{\text{\# Identified from application data}} \times \frac{\kappa(v)}{G_t(v)^2},$$

and $G_t(v)$ is observed from our data on applications and recipients.

Once we have identified $f_{\nu_t}(v)$, $h(m_t(v))$, and $\beta_k^b(v)$, $\beta_k^w(v)$, or $\kappa(v)$, we can quantify Δ_{kt}^b, Δ_{kt}^w, or Δ_t^G from the observed time changes in the k'th benefit shifter z_{kt}^b, in the k'th work shifter z_{kt}^w, or in the award function $G_t(v)$. As shown in Section 6, this allows us to construct logically coherent predictions of counterfactual trends in SSDI application rates. But first, we present the quasi-experimental estimates that pin down $\beta_k^b(v)$, $\beta_k^w(v)$, or $\kappa(v)$.

5.2 Quasi-experimental estimates

We now present a set of quasi-experimental estimates on SSDI application of plausibly exogenous changes in a wide range of factors that could affect incentives to apply for SSDI (i.e., $\beta_k^b(v)$, $\beta_k^w(v)$, or $\kappa(v)$ from the model above). Table 3 summarizes the factors we consider.

The inverse relationship between SSDI application and employment found in Section 4 suggests the following possibilities:

1. **Higher labor demand → lower SSDI.** Higher demand for less-skilled workers could

have increased their employment rates and diverted them from SSDI.[22]

2. **Lower SSDI due to lower award rates → higher employment.** Lower SSDI award rates at the appeals level, attributed in part to the retraining of judges, may have discouraged applications.[23]

3. **Lower SSDI due to higher administrative burden → higher employment.** Changes in wait times, field office service, representative compensation, or information could have discouraged applications.[24] The political discussion around social welfare benefits in the run-up to the 2016 election could have also discouraged applications.

4. **Third factor (e.g., better health) → lower SSDI, higher employment.** Reductions in pollution and smoking could have improved public health and reduced SSDI application rates. More effective treatments for mental health conditions and more availability of treatment could have also reduced SSDI applications.

5. **Higher mortality of potential applicants → lower SSDI, higher employment.** By increasing the mortality of potential applicants, the opioid crisis could have both reduced demand for SSDI and mechanically increased employment rates (if those potential applicants would not have worked).

6. **More generous benefits (inconsistent with work patterns).** If other benefits became more generous, people may have substituted from SSDI to those benefits, but then we would not expect to see higher employment rates.[25]

7. **Population aging (inconsistent with work patterns).** The aging of the population could have moved people from the SSDI program to the retirement program, but then we would not expect to see higher employment rates.

Table 3 summarizes the quasi-experimental evidence on all of these factors. We first determine whether the relevant factor changed during the period of the application decline (2010–19). We then use either novel evidence or existing evidence to determine the responsiveness of SSDI applications to that factor. By combining the change in the factor with the responsiveness of applications to that factor, we determine to what extent that factor could have contributed to the decline in SSDI applications.

[22]Glenn and Goss (2018); Ruffing (2023); Center on Budget and Policy Priorities (2025); Liu and Quinby (2023).

[23]See Glenn and Goss (2018); Social Security Administration (2019); Ruffing (2020, 2023); Center on Budget and Policy Priorities (2025); Liu and Quinby (2023); Mohamed et al. (2024).

[24]See Glenn and Goss (2018); Armour (2018); Ruffing (2023); Center on Budget and Policy Priorities (2025); Mohamed et al. (2024).

[25]See Glenn and Goss (2018); Ruffing (2023); Center on Budget and Policy Priorities (2025).

Table 3: Evaluating hypotheses for the decline in SSDI applications from 2010 to 2019

	Change in factor 2010–19?	Empirical strategy for response of SSDI applications	Explanatory power?	See for details
Consistent with SSDI application and work patterns				
1) Labor market conditions				
Labor demand	Yes, long recovery	Use Autor and Duggan (2003) shift-share design	Likely large	Section 5.2.1
Remote work	Yes, more remote jobs	Heterogeneity based on Dingel and Neiman (2020)	Little, if any	Section 5.2.1
Minimum wage	Yes, state increases	Heterogeneity by local minimum wage	Little, if any	Section 5.2.1
2) Award rates				
Award rates	Yes, ALJ reform	DiD using baseline ALJ award rate	Little, if any	Section 5.2.2
3) Admin burden				
Field office closings	Yes, several closings	Existing evidence: Deshpande and Li (2019)	Little, if any	Section 5.2.3
Processing times	No		Little, if any	Section 5.2.3
Legal representation	Yes, ALJ reforms	DiD using baseline legal representation	Little, if any	Section 5.2.3
Social Security mailings	No, only stopped 2011–14		Little, if any	Section 5.2.3
Stigma	Possibly, political discussion	Heterogeneity by political leaning	Little, if any	Section 5.2.3
4) Health				
Pollution	Yes, 1970 Clean Air Act	Heterogeneity based on Isen et al. (2017)	Little, if any	Section 5.2.4
Smoking	No		Little, if any	Section 5.2.4
Mental health treatment	Yes, higher take-up	Existing evidence + heterogeneity by diagnosis	Potentially some	Section 5.2.4
5) Mortality				
Opioid crisis	Yes	Heterogeneity by opioid death rate	Little, if any	Section 5.2.5
Inconsistent with SSDI application and work patterns				
6) Other benefits				
SNAP/TANF/UI	No		Little, if any	Section 5.2.6
Other cash benefits	Yes, VA Disability	Heterogeneity by veteran status	Little, if any	Section 5.2.6
Health insurance	Yes, ACA	Existing evidence: many sources (see text)	Little, if any	Section 5.2.6
Health care access	Yes, CHCs	Existing evidence: Anstreicher and Strand (2025)	Little, if any	Section 5.2.6
7) Population aging				
Population aging	No, working-age population has shifted to highest-DI-receipt ages		Little, if any	Section 5.2.7

Notes: The first column lists various factors that could potentially have contributed to the decline in SSDI applications from 2010 to 2019. The second column summarizes the evidence on whether that factor changed during the period of the application decline (2010–19). The third column describes how we determine the responsiveness of SSDI applications to that factor, using either new evidence (either descriptive or quasi-experimental) or existing evidence. The fourth column describes, based on the combination of the second and third columns, whether the factor could potentially explain some part of the decline in SSDI applications, which previews our analysis in Section 6 about whether this factor is likely to be important in explaining the decline. See text for details.

5.2.1 Improvement in the labor market for less-skilled workers (Higher labor demand → lower SSDI)

In a widely cited paper, Autor and Duggan (2003) conclude that SSDI application rates are sensitive to changes in demand for less-skilled workers, especially after the 1984 expansion of medical eligibility. Motivated and guided by their finding and research design, we examine whether improvements in the labor market conditions of less-skilled men (i.e., with no college education) from 2010 to 2019 could have reduced SSDI application rates.

The time series of unemployment and non-employment rates suggests that this could indeed be the case. As shown in Figure 9, both the unemployment rates and the employment-to-population ratio (the share of the working-age population who are currently working) track SSDI application rates closely over time for less-skilled men (see Appendix Figure A.9 for all adults). The steady and prolonged recovery from the Great Recession is no exception. As the labor market recovered, SSDI application rates fell. We also find in the cross-section that places with larger declines in local unemployment rates contributed disproportionately to declines in SSDI application rates from 2010 to 2019 (see Appendix Table H.16).

Figure 9: SSDI applications track unemployment and non-employment rates over time

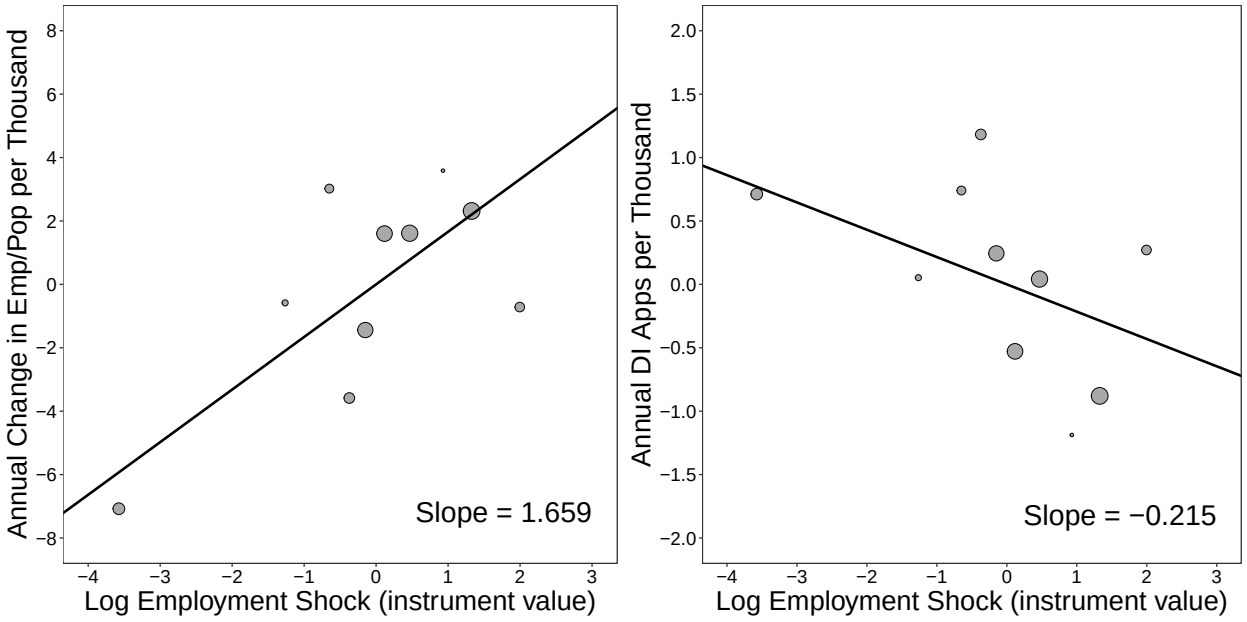


(a) Applications and unemployment, less-skilled men ages 18-66

(b) Applications and non-employment, less-skilled men ages 18-66

Notes: Figure plots SSDI applications, unemployment rates, and non-employment rates over time for men ages 18-66 years with no college education. SSDI applications are per thousand working-age adults. Employment measures (dashed lines) are plotted against the secondary axis in panel. Non-employment is reported as a percent of the working-age population, and unemployment as a percent of the labor force. Both are measured using monthly CPS data from IPUMS. Time series correlations of applications with non-employment and unemployment are 0.53 and 0.73, respectively. The correlation between non-employment and unemployment is 0.65. See Appendix Figure A.9 for analogous graphs using the full adult population.

Figure 10: Local labor market improvements positively correlated with employment and negatively correlated with SSDI application rates



(a) Employment and Log Employment Shock (b) Applications and Log Employment Shock

Notes: The data used to construct these plots is composed of non-overlapping three-year intervals for each state. Intervals used to construct the data are: (2013,2016], and (2016,2019]. All variables have been residualized with respect to the controls included in equation (20): time dummies and linear controls for annualized changes in population shares by age and gender, age being split into 3 bins: 18-39, 40-54, and 55-66. Both panels present bin scatter plots, constructed by assigning state-interval observations into 10 equally sized groups based on percentiles of the explanatory variable (log employment shock). Each circle shows the population-weighted average outcome (change in employment or SSDI applications) and average log employment shock within each bin. The size of each circle reflects the total population in that bin.

There is also a strong spatial association between SSDI application rates and changes in local employment and labor market conditions. To show this, we follow Autor and Duggan (2003). We first use CPS data to re-construct their measure of state-level labor demand shifts, defined as a weighted sum of national industry employment changes of less-skilled men (excluding own state employment) projected onto state industry composition. Next, Figure 10 plots these measures of labor demand shifts against changes in the employment of less-skilled men (10a) and the SSDI application rates (10b) of less-skilled men in each state. The key insight from Figure 10 is that the Autor and Duggan (2003) measures of labor market shifts continue to co-vary strongly with employment rates and SSDI application rates at the state level in more recent years.

To quantify and economically interpret the relationships in Figure 10, we estimate the following 2SLS regression model:

$$(\text{Apps per Capita})_{j,\tau} = \alpha_0 + \lambda_{t(\tau)}\Delta(\text{Emp per Capita})_{j,\tau} + \beta_0' X_{j,\tau} + \varepsilon_{j,\tau} \tag{20}$$

with the corresponding first-stage equation:

$$\Delta(\text{Emp per Capita})_{j,\tau} = \alpha_1 + \gamma_{t(\tau)}\hat{\eta}_{j,\tau} + \beta_1' X_{j,\tau} + \vartheta_{j,\tau} \tag{21}$$

where X denotes a set of calendar year fixed effects and linear demographic controls (changes in population shares by sex and age group: 18-39, 40-54, 55-66), and τ represents an interval of three years. The dependent variable, $(\text{Apps per Capita})_{j,\tau}$, is the average annual application rate of less-skilled men in state j over the three-year interval τ. We include two non-overlapping three-year intervals: 2013-2016 and 2016-2019 The endogenous variable, $\Delta(\text{Emp per Capita})_{j,\tau}$, is the change in the employment-to-population ratio for less-skilled men over τ (annualized by dividing this change by 3). Our instrument, $\hat{\eta}_{j,t}$, is the exact same measure of state-level labor demand shifts used by Autor and Duggan (2003).

Appendix Table C.1 presents the 2SLS estimates.[26] Our main 2SLS estimate for less-skilled men is -0.13 (with a standard error of 0.16): in a given three-year interval, an annualized 1 percentage point growth in the employment-to-population rate yielded a 0.13 percentage point decline in annual application rates. This average effect masks heterogeneity across individual types, especially with respect to age. For example, effects for older high school graduates (-0.33 with a standard error of 0.15) are several times larger than effects for younger high school graduates (-0.05 with a standard error of 0.07).

While the point estimates indicate substantial effects of employment on DI application, the standard errors are relatively large.[27] To gain precision, we consider two alternative specifications. Both of these specifications lead to larger and highly significant estimates, suggesting our original specification gives a conservative estimate of the effects of employment on DI application. Neither is significantly different from our main estimate.

The first additional specification we consider is to use additional years of data in the estimation of our 2SLS model in equations (20) and (21). Specifically, we include every non-overlapping three year-difference from 1992 to 2019 (i.e., 1992-1995, 1995-1998, ..., 2013-2016, 2016-2019).[28] The reason we start in 1992 is that it begins the earliest three-year

[26] An implication of the model is that, conditional on type, a shift in the payoff of working will not change applicant award rates. Appendix Table C.2 shows that we find no evidence that the shifter affects award rates among applicants.

[27] The main estimates of Autor and Duggan (2003) also have relatively large standard errors. Focusing on the point estimate, they conclude that improvements in the labor market after 1984 is a key driver of the rise in DI applications.

[28] Our findings are similar if we instead include all overlapping three-year intervals (i.e., 1992-1995, 1993-1996, ..., 2015-2018, 2016-2019).

interval for which we can observe application data by education and geography. We then find a 2SLS estimate of -0.43 with a standard error of 0.15, which is significant at even the 1 percent confidence level.

Another specification that we consider is to use the exact same data and regression model as Autor and Duggan (2003). The key difference relative to our original specification is that Autor and Duggan (2003) only observe DI application at the state level, and therefore, their outcome variable is DI application rates in the population at large, and not for the less-skilled (for whom they measure the endogenous variable, the employment rate). When re-estimating their baseline specification from 1984 to 2019, we find a 2SLS estimate of -0.37 with a standard error of 0.09. This estimate is also significant at the 1 percent level.

In Section 6, we show that both the estimates from our original specification and those from the alternative specifications are large enough to explain much of the decline in SSDI applications from 2010 to 2019. In fact, even the lower end of the 95-percent confidence interval of the alternative estimates suggests that changes in employment are a key driver of the decline in SSDI applications from 2010 to 2019.

Contribution of other labor market factors, like remote work and minimum wage. Another labor-market-related explanation for the decline in SSDI applications is that improvements in technology enabled more workers to work remotely, which could have increased the labor *supply* of workers with disabilities and thereby reduced SSDI applications.[29] We find no heterogeneity in the decline in SSDI application by the Dingel and Neiman (2020) measure of feasibility of remote work across industries (see Appendix Table H.26).

Under some circumstances (like firm monopsony power), higher minimum wages could increase employment and reduce SSDI applications. However, we do not find a correlation between local minimum wage and SSDI application decline (see Appendix Table H.19).

Previous economic recoveries. Prior economic recoveries in recent SSDI history occurred from 1993 to 2000, and from 2004 to 2006. A natural question is why these prior recoveries did not produce declines in SSDI applications. In the case of the 1993–2000 recovery, one obvious possibility is that there were countervailing forces pushing up applications: medical eligibility criteria for SSDI had been expanded in 1984, and the cohorts that aged immediately after that decision applied in greater numbers in response to that policy change. In addition, Autor and Duggan (2003) document stagnating real wages at the bottom of the earnings distribution during this time. The 2004–2006 economic recovery—just three years

[29] Bloom et al. (2026) estimate that the work-from-home revolution accounts for 80% of the post-pandemic increase in employment among people with disabilities.

long, compared to 10 years for 2010–2019—may simply have been too short to produce meaningful declines in SSDI applications.

5.2.2 Decline in award rates on appeal (Lower SSDI → higher employment)

One of the leading hypotheses for the SSDI application decline is that the decline in award rates at the appeals level has discouraged potential applicants from applying for SSDI (Glenn and Goss, 2018; Social Security Administration, 2019; Ruffing, 2023; Center on Budget and Policy Priorities, 2025; Liu and Quinby, 2023; Mohamed et al., 2024). Policy changes starting in 2008, including new training and guidelines for administrative law judges, reduced award rates at the appeals level (Hoynes et al., 2024), which could have subsequently discouraged applications and increased labor market earnings.[30]

We estimate the effect of award rates on SSDI applications. Since ALJs have specific geographic jurisdictions, we use geographic variation in award rates at the appeals level (specifically, the likelihood of award conditional on being rejected at the initial level) prior to 2008 to estimate the effect of a change in award rates on SSDI applications. We first establish that areas with higher allowance rates at the appeals level prior to 2008 experienced larger decreases in award rates after 2008, and then estimate the effect on SSDI applications in future years.

We estimate the following difference-in-differences specification:

$$Y_{it} = \alpha_i + \gamma_t + \sum_t \beta_t(\text{BaseAppealAllow}_i \times \text{D}_t) + \delta X_{it} + \epsilon_{it} \tag{22}$$

where Y_{it} is an outcome like appeals award rate (first stage) or future applications (reduced form) for county i in year t; BaseAppealAllow_i is the baseline appeals allowance rate (share of applications rejected at the initial level that are allowed on appeal in the three years prior to the reform); D_t are year indicators; α_i are county fixed effects; γ_t are year fixed effects; and X_{it} are time-varying characteristics of counties (e.g., unemployment rate). We weight counties by the number of applications in the three years prior to reform.[31]

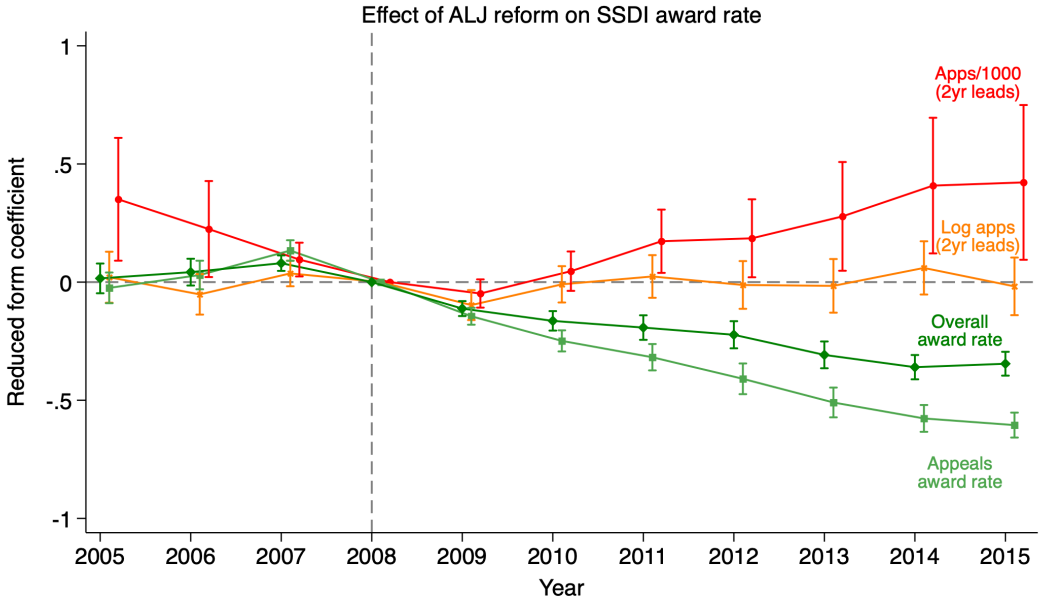
The main results are shown in Figure 11. We find that the ALJ reform reduces appeals award rates by 44 pp more, and overall allowance rates by 28 pp more, for (hypothetical)

[30] Papers from across the developed world show that DI denial or removal increases labor market earnings. See, e.g., Chen and van der Klaauw (2008); Von Wachter, Song and Manchester (2011); Maestas, Mullen and Strand (2013); Borghans, Gielen and Luttmer (2014); French and Song (2014); Moore (2015); Deshpande (2016); Autor, Kostøl, Mogstad and Setzler (2019); Haller, Staubli and Zweimüller (2024).

[31] We construct the number of SSDI applications in a county-year using the F831 file, which includes ZIP code. We construct one, two, and three year leads of applications since application responses to award rates may take time to manifest. To construct baseline appeals award rates (and overall award rates), we link the F831 file to payment data from the PHUS.

places with a 100% baseline allowance rate relative to (hypothetical) places with a 0% baseline allowance rate over the next 7 years. However, two-year application leads do not decline in areas with falling award rates (see Appendix Table A.5 for robustness checks).

Figure 11: SSDI applications do not fall more in areas more affected by ALJ reform



Notes: Figure plots estimates of the effect of a county having a higher baseline appeals award rate in 2008 (i.e., β_t from equation (22)) on the following outcomes: SSDI appeals award rate in the county (light green), SSDI overall award rate in the county (dark green), number of SSDI applications two years later (divided by 100) in the county (red), and log SSDI applications two years later in the county (orange). The sample is constructed from the F831 file linked to the Payment History Update System. Appeals award rate is measured as the fraction of SSDI applications rejected at the initial level in the F831 that are eventually awarded SSDI benefits. Overall award rate is measured as the fraction of SSDI applications in the F831 that are eventually awarded SSDI benefits.

5.2.3 Administrative burden of applying (Lower SSDI → higher employment)

We consider several potential explanations under the broad umbrella of administrative burden, but find that none can explain a large share of the decline in applications.

Field office closings. Deshpande and Li (2019) estimate that Social Security field office closings reduce SSDI applications and awards substantially in neighboring communities. However, given the limited number of closings during this time, we find that they could only explain 4.4% of the decline in SSDI applications. See Appendix Section B.2.1 for details on the exact calculation.

Processing times. Processing times have not risen systematically in recent years.[32]

Legal representation. Hoynes, Maestas and Strand (2026) document a decline in legal representation of SSDI applicants at the appeals level following the ALJ reforms discussed above, since lower appeals award rates reduced lawyer fees. They find that this change increased representation at the initial application level. We show in Figure 11 that the ALJ reforms had no effect on application by comparing areas with high versus low baseline appeals award rates. We also show in Appendix Figure A.10 that applications did not decline more for areas with high baseline levels of legal representation versus areas with low baseline levels following the ALJ reform.

End of Social Security statement mailings. SSA stopped mailing Social Security statements in 2011, restored the mailings in 2014, and then limited them again in 2017. These changes could have reduced worker awareness of SSDI benefits and their eligibility. However, the persistence of the decline in applications from 2010 to the present suggests that this change is not the primary explanation.

Stigma. The stigma of disability receipt could have changed over this time period, given the spotlight on the safety net in the national political debate. While we do not have a direct measure of stigma, we find no substantial heterogeneity in the decline in applications by state political leaning or by urban versus rural areas (see Appendix Tables H.12 and H.14).

5.2.4 Changes in health (Third factor → lower SSDI, higher employment)

Improvements in general health. Since our administrative data do not include health measures for the full population, we use the National Health Interview Survey to determine whether population health improved during the period of SSDI application decline. We find that neither overall health, nor rates of mental and musculoskeletal conditions specifically, has improved for low-to-middle-skilled men (see Appendix Figures A.11 and A.12).

Public health measures to reduce pollution and smoking. We examine whether public health measures to reduce pollution and smoking rates could have reduced SSDI applications.

[32]The SSA reports average processing times for all cases nationwide only starting from 2014 (`https://www.ssa.gov/ssa-performance/disability-appeals-time`). Kearney, Price and Wilson (2021) find that, in areas where employment declined sharply during the Great Recession, areas assigned to SSA offices with longer appeals processing times experience lower SSDI enrollment, but the effect is modest.

- Pollution abatement: Isen et al. (2017) find that reductions in childhood pollution exposure due to the Clean Air Act of 1970 are associated with higher labor force participation in adulthood. Using data provided by Isen et al. (2017), we classify individuals by the pollution abatement requirements in the Clean Air Act for their place of birth. We find no heterogeneity in the decline in SSDI application based on this measure (see Appendix Section B.2.2 for more details on the analysis and Appendix Section G.3 for results).

- Decline in smoking: Smoking rates among men declined steadily from around 70% for the 1925 birth cohort to around 35% for the 1965 birth cohort, before plateauing for more recent birth cohorts (Holford, Levy, McKay, Clarke, Racine, Mez, Land, Jeon and Feuer, 2014). Men in birth cohorts between 1925 and 1965 would have turned 50 years old between 1975 and 2015. To the extent that lower smoking rates reduce SSDI applications, this effect would have started well before 2010. For this reason, lower smoking rates alone probably cannot explain the reversal of SSDI application growth in 2010, even though SSDI application levels may well have been higher in the absence of smoking reductions. We also find no heterogeneity in the application decline by county smoking rates (see Appendix Table H.24).[33]

Improvements in mental health treatment. Using data from the National Health and Nutrition Examination Survey (NHANES), we find that while rates of severe depression have not fallen in the working-age population or among low-skilled men, the use of antidepressants and antipsychotics has increased since 2010 (see Appendix Figures A.13 and A.14). Some papers estimate sizable increases in employment and/or reductions in disability benefits take-up from mental health treatment (Biasi, Dahl and Moser, 2027; Shapiro, 2022; Bütikofer, Cronin and Skira, 2020), while others estimate zero effects (Laird and Nielsen, 2016; Serena, 2024). Given the wide range of estimates, we cannot rule out improvements in mental health treatment as a driver of the decline in SSDI applications. It is unlikely to be the only or main driver, however, since the decline happens across all types of conditions, as shown in Figure 6.

[33]Using a sibling fixed effects model, Bengtsson and Nilsson (2018) find that smoking increases the likelihood of receiving disability benefits in Sweden by 4.2pp. Combining this estimate with the approximately 10pp reduction in male smoking rates between the 1955 and 1965 birth cohorts (Holford et al., 2014), we would expect smoking to reduce SSDI receipt rates among men by 0.4pp across these cohorts. This reduction is 20% of the actual reduction in SSDI receipt rates among men across these cohorts.

5.2.5 Opioid crisis (Higher mortality → lower SSDI, higher employment)

While drug addiction itself is no longer a qualifying condition for SSDI or SSI receipt, the opioid crisis could have led to comorbidities that do qualify individuals for SSDI receipt. We would expect such comorbidities to *increase* SSDI application and enrollment. On the other hand, if the opioid crisis resulted in the death or incapacitation of people who would have applied for SSDI, it could explain some of the decline in SSDI application. We find no meaningful heterogeneity in the decline in SSDI applications across counties by the severity of the opioid crisis (see Appendix Table H.21).[34]

5.2.6 More generous other benefits (inconsistent with work patterns)

There was no increase in SNAP, TANF, or UI benefit amount or enrollment between 2010 and 2019 (see Appendix Figures B.1, B.2, and B.3). However, there were major changes in health insurance coverage and Veterans Disability Benefits during this period.

Affordable Care Act expansion of health insurance coverage. The Affordable Care Act of 2010 expanded access to health insurance to lower- and middle-income adults through Medicaid (in certain states) and the exchanges, starting in 2014. If potential SSDI applicants would have applied primarily to get health insurance coverage, then the ACA expansions could have contributed to the SSDI application decline. We investigate this hypothesis in several ways. First, we examine the correlation between increases in health insurance coverage and changes in SSDI applications (Appendix Figure B.4). We find that larger increases in health insurance coverage are associated with increases in SSDI applications, which is inconsistent with the hypothesis that health insurance coverage reduced demand for SSDI. Second, we study whether the decline in SSDI applications is concentrated in states that expanded Medicaid coverage (Appendix Table H.13). We find no heterogeneity in the SSDI application decline by state Medicaid expansion status. Finally, we review the large existing literature on the causal evidence on Medicaid expansions, with more details in Appendix Section B.2.3. Consistent with our correlational analysis, these papers find that Medicaid expansions had either no effect or a positive effect on SSDI and SSI applications.[35]

[34] We use the CDC Wonder Database to measure deaths from opioids from 2010–2019, using the following ICD-10 codes: T40, T40.1, T40.2, T40.3, T40.4, T40.6. In the statistical decomposition, changes in mortality rates of SSDI recipients do not account for any of the decline in SSDI enrollment rates (Figure 3b).

[35] We review the following papers: Baicker, Finkelstein, Song and Taubman (2014); Maestas, Mullen and Strand (2014); Burns and Dague (2017); Soni, Burns, Dague and Simon (2017); Anand, Hyde, Colby and O'Leary (2019); Schmidt, Shore-Sheppard and Watson (2020); Heim, Lurie, Mullen and Simon (2021); Ne'eman and Maestas (2023a); Staiger, Helfer and Van Parys (2024). See Appendix Section B.2.3 for details.

Veterans Disability Compensation. Enrollment in Veterans Disability Compensation (VDC) has increased from 3.2 million in 2010 to 5.7 million in 2023 (U.S. Department of Veterans Affairs, 2010, 2023). While there is no restriction on receiving both VDC and SSDI benefits, it is possible that higher VDC receipt could reduce SSDI application and enrollment through an income effect. To evaluate this hypothesis, we identify veterans in our data based on military EINs. However, we find that veterans have a disproportionately small decline in SSDI applications relative to their size (see Appendix Table G.26).

On the other hand, the share of older men who are veterans has decreased substantially from 2000 to 2020 (Chan, Duggan and Guo, 2021). We investigate whether a decline in the population share of veterans could have contributed to the decline in SSDI enrollment. We redo the decomposition exercise in Table 2 including veteran status (and thus the interaction between veteran status and age), with results in Appendix Table G.23. Including veteran status as a characteristic of the decomposition does not change the main results. See Appendix Section B.2.3 for more details.

5.2.7 Population aging (inconsistent with work patterns)

From Table 2, population aging does not play a major role in the decline in SSDI enrollment among the working-age population. This is likely for two reasons: First, population aging meant that the working-age population was on average older in 2019 than in 2010, and older workers have higher SSDI rates (see Appendix Figure A.15). Second, we analyze SSDI enrollment rates in the working-age population, which by definition does not include people above retirement age.

6 Counterfactual trends in SSDI applications

We now combine the causal estimates above with the model to construct predictions of counterfactual trends in SSDI applications.

6.1 Constructing counterfactual SSDI trends

Recall the model from Section 5.1. To construct the counterfactual SSDI trend in the absence of, say, the observed time changes in the k'th work shifter z^w_{kt}, it could be tempting to simply set $\Delta^w_{kt} = 0$ for all $t_0 \leq t \leq t_1 - 1$. However, this would be misleading, as changes in z^w_{kt} from one period to another not only affect application shares directly through Δ^w_{kt}, but also change how many individuals are close to the margin of application in the future, as it changes future values of m_t. We show how to account for these considerations when

constructing counterfactual application rates in the absence of observed changes in the labor demand shifter (z^w_{kt}) that we study in sub-section 5.2.1, and the award rate shifter (G_t) that we study in sub-section 5.2.2.

Changes in the work shifter. To account for both direct and indirect channels, we introduce the counterfactual leniency-adjusted benefit of applying, which is recursively defined by

$$\tilde{m}^w_{t+1}(v;k) = \tilde{m}^w_t(v;k) + m_{t+1}(v) - m_t(v) + \underbrace{\beta^w_k(v)\left(z^w_{k,t+1} - z^w_{k,t}\right)}_{\text{period } t\text{-effect of no change in } z^w_k}$$

with initial condition $\tilde{m}^w_{t_0}(v;k) = m_{t_0}(v)$. $\tilde{m}^w_{t+1}(v;k)$ measures the leniency-adjusted benefit of applying at time t had $z^w_{k,t}$ remained fixed from t_0 onwards. It is straightforward to solve the difference equation and obtain

$$\tilde{m}^w_t(v;k) = m_t(v) + \beta^w_k(v)\left(z^w_{k,t} - z^w_{k,t_0}\right). \tag{23}$$

This sequence also generates a sequence of counterfactual type-specific application rates $H(\tilde{m}^w_t(v;k))$ where we use skill groups as types.

The counterfactual *aggregate* trend in SSDI applications in the absence of the observed time changes in the k'th work shifter z^w_{kt} can then be defined by:

$$\mathbb{P}(A^w_{t_1}(k) = 1) = \int H(\tilde{m}^w_t(v;k)) f_{v_t}(v) dv \tag{24}$$

where $A^w_{t_1}(k)$ denotes application decision at time t_1 if z^w_{kl} is set equal to $z^w_{kt_0}$ from period t_0 to t_1. As explained in sub-section 5.1, we can compute this counterfactual by combining the quasi-experimental estimates in sub-section 5.2.1 with yearly micro data on SSDI applications to measure $m_t(v)$ and the observed time changes in z^w_{kt}.

Changes in the award function. We can consider the effect of turning off changes in the award function G_t in a similar manner. Define a counterfactual (setting $G_{t+1} = G_t$) sequence of leniency-adjusted benefits of applying by:

$$\tilde{m}^G_{t+1}(v) = \tilde{m}^G_t(v) + m_{t+1}(v) - m_t(v) + \underbrace{\left(\frac{\kappa(v)}{G_{t+1}(v)} - \frac{\kappa(v)}{G_t(v)}\right)}_{\text{period } t\text{-effect of no change in } G},$$

with initial condition $\tilde{m}^G_{t_0}(v) = m_{t_0}(v)$. This gives

$$\tilde{m}^G_t(v) = m_t(v) + \kappa(v)\left(\frac{1}{G_t(v)} - \frac{1}{G_{t_0}(v)}\right).$$

Again, we can write the counterfactual *aggregate* trend in SSDI applications as

$$\mathbb{P}(A^G_{t_1} = 1) = \int H(\tilde{m}^G_t(v)) f_{v_t}(v) dv,$$

where A^G_t denotes application decision at time t if G_t remained constant from t_0 to t_1. As explained in sub-section 5.1, we can compute this counterfactual by combining the quasi-experimental estimates in sub-section 5.2.2 with yearly micro data on SSDI applications to measure $m_t(v)$ and the observed time changes in G_t.

6.2 Counterfactual results

In Figure 12, we apply equations (23) and (24) to compute application rates under six counterfactual scenarios: (a) the employment-to-population ratio for less-skilled men had remained at its 2010 value, (b) the SSDI award rate had been fixed at pre-recession levels, (c) the number of SSA field offices had been fixed at pre-recession levels, (d) Medicaid access had not been expanded under the ACA, (e) the population had not aged relative to pre-recession levels, and (f) the veteran population share and application rate had been fixed at pre-recession levels. Appendix Section C includes details on the calculations of these counterfactuals.

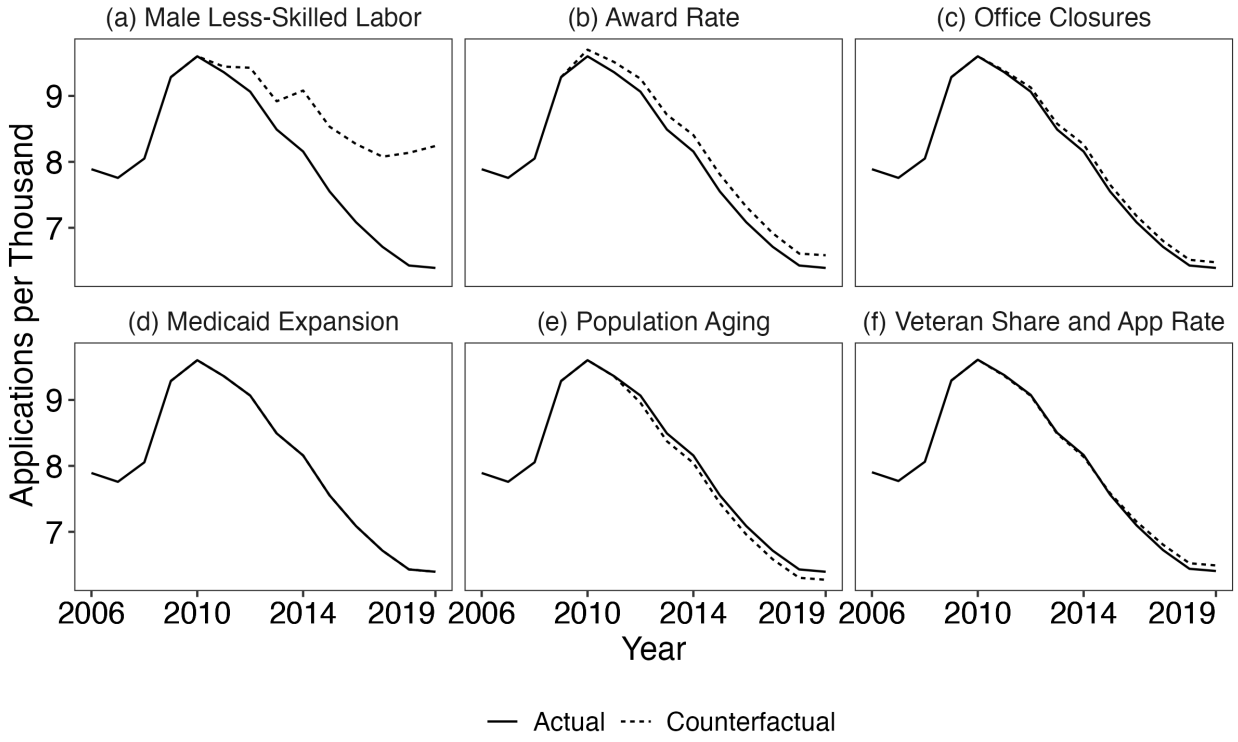
Fixing male less-skilled labor market conditions—allowing all else to vary over time—produces a counterfactual application rate in 2019 that is close to its baseline 2010 value. This exercise suggests that in 2019, rising employment among less-skilled men can explain 57 percent of the decline in the application rate relative to 2010. Appendix Figure C.1 shows that labor market conditions for less-skilled men entirely explain the decline in application rates among men in particular. In contrast, the other counterfactuals in Figure 12 suggest that award rates, office closures, Medicaid expansions, population aging, and changes in veteran composition or benefits cannot explain much of the decline.

We translate these SSDI application counterfactuals into counterfactual changes in enrollment, using back-of-the-envelope calculations with actual award rates by type and over time. Cumulatively from 2010 to 2019, improvements in labor market conditions for less-skilled men reduced SSDI enrollment by 781,000, or 0.39 percent of the 2019 working-age population. By comparison, the total decline in SSDI enrollment rate in the working-age

population, relative to the counterfactual, is 1.04 pp (Table 2).

A possible concern with this counterfactual analysis is that the 2SLS estimate has relatively large standard errors. We addressed this concern in Section 5.2.1 by estimating two alternative specifications using additional years of data. The resulting estimates then became larger and significant. In Appendix Figure C.2, we plot confidence intervals on counterfactuals using our original estimate and these alternative specifications. We find that even the lower end of the 95-percent confidence interval of the alternative estimates suggests that changes in employment rates are a key driver of the decline in SSDI applications from 2010 to 2019.

Figure 12: Labor demand for less-skilled men explains substantial share of SSDI application decline



Notes: Figure presents counterfactuals based on the method discussed in Section 6. When the "counterfactual" line is not visible, it is on top of the "actual" line. For panel (d) type is state of residence. For panel (e), type is age (18-24, 25-30, 31-40, 41-50, 51-64, 65-66). For panel (f), type is veteran status. For all other panels, type is defined by a combination of sex, educational attainment, and age (18-40 vs. 41-66). In panel (e) we fix the population age composition as observed in 2010. In panel (f), we fix both the veteran population share and application rates among veterans. For more details on estimation, see Appendix Section C.

7 Conclusion

The increase in SSDI enrollment in the 1990s and 2000s sparked significant policy debate and concern about the fairness of the social safety net and the consequences of a declining workforce. Yet the subsequent drop in SSDI applications and enrollment starting in 2010 has received much less attention and prompted little reconsideration about those previous concerns.

We find that the recent decline in SSDI enrollment is driven primarily by declines in applications, and secondarily by declines in awards, with minimal contribution from population composition, work-history eligibility, or exit. The decline in SSDI applications comes disproportionately from low-to-middle skilled men and disproportionately from applicants who have relatively severe conditions and are strongly attached to the labor force. Potential applicants appear to switch from applying for SSDI to working. Low- and middle-skilled men with health limitations during this time have historically high earnings levels and live with spouses with historically high earnings levels. In this population, income from SSDI is replaced with income from work.

We find that the most likely explanation for the decline in applications is improved work opportunities for low- and middle-skilled men with health limitations. Strong economic opportunities at the bottom of the earnings distribution appear to have pulled these men into the labor force and diverted them, at least temporarily, from SSDI. We find evidence that is inconsistent with other explanations for the decline in SSDI applications, including lower award rates at the appeals level, higher administrative burden of applying, improved population health, and increased generosity of other programs.

The evidence that better employment opportunities drive the SSDI decline prompts two questions. First, will the marginal non-applicants apply for SSDI benefits in the event of an economic downturn? Perhaps so, but the prolonged economic recovery from 2010 to the present means that many potential applicants have already hit retirement age and have therefore been permanently diverted from SSDI. Second, can policies that improve employment opportunities divert more people from SSDI? It is unclear whether temporary jobs or training programs can provide the same opportunities to low- and middle-skilled workers that the economic recovery in the 2010s did. To the extent that they can, however, successful efforts to divert individuals from SSDI could amount to substantial cost savings and higher long-term labor force participation rates.

References

Aeppli, Clem and Nathan Wilmers, "Rapid wage growth at the bottom has offset rising US inequality," *PNAS*, 2022, *119* (42), 1–7. 32

Anand, Priyanka, Jody Schimmel Hyde, Maggie Colby, and Paul O'Leary, "The Impact of Affordable Care Act Medicaid Expansions on Applications to Federal Disability Programs," *Forum for Health Economics and Policy*, December 2019, *21* (2). Publisher: De Gruyter Section: Forum for Health Economics & Policy. 49, 23, 33

Anstreicher, Garrett, "Does increasing health care access reduce disability insurance caseloads? Evidence from the rural United States," *Health Economics*, July 2021, *30* (4), 786–802. 26

_ **and Alexander Strand**, "Health Care Access and Disability Insurance Applications," Working Paper 2025. 40, 26

Armour, Philip, "The role of information in disability insurance application: an analysis of the social security statement phase-in," *American Economic Journal: Economic Policy*, 2018, *10* (3), 1–41. 2, 39

Autor, David, Andreas Kostøl, Magne Mogstad, and Bradley Setzler, "Disability Benefits, Consumption Insurance, and Household Labor Supply," *American Economic Review*, July 2019, *109* (7), 2613–54. 45

Autor, David H. and Mark G. Duggan, "The Rise in the Disability Rolls and the Decline in Unemployment," *The Quarterly Journal of Economics*, February 2003, *118* (1), 157–206. 1, 5, 6, 40, 41, 42, 43, 44, 28, 30

_ **and** _ , "The Growth in the Social Security Disability Rolls: A Fiscal Crisis Unfolding," *Journal of Economic Perspectives*, September 2006, *20* (3), 71–96. 1, 6

Baicker, Katherine, Amy Finkelstein, Jae Song, and Sarah Taubman, "The Impact of Medicaid on Labor Market Activity and Program Participation: Evidence from the Oregon Health Insurance Experiment," *American Economic Review*, 2014, *104* (5), 322–328. 49, 23, 33

Bengtsson, Tommy and Anton Nilsson, "Smoking and early retirement due to chronic disability," *Economics and Human Biology*, 2018, *29*, 31–41. 48

Biasi, Barbara, Michael S. Dahl, and Petra Moser, "Career Effects of Mental Health: Evidence from an Innovation in Treating Bipolar Disorder," *Journal of Political Economy Microeconomics*, 2027. Forthcoming. 48

Blinder, Alan S., "Wage Discrimination: Reduced Form and Structural Estimates," *The Journal of Human Resources*, 1973, *8* (4), 436–455. 2

Bloom, Nicholas, Gordon B Dahl, and Dan-Olof Rooth, "Work from Home and Disability Employment," *American Economic Review: Insights*, 2026, *Forthcoming.* 6, 44

Borghans, Lex, Anne C. Gielen, and Erzo F. P. Luttmer, "Social Support Substitution and the Earnings Rebound: Evidence from a Regression Discontinuity in Disability Insurance Reform," *American Economic Journal: Economic Policy*, November 2014, *6* (4), 34–70. 45

Burns, Marguerite and Laura Dague, "The effect of expanding Medicaid eligibility on Supplemental Security Income program participation," *Journal of Public Economics*, May 2017, *149*, 20–34. 49, 23

Bütikofer, Aline, Christopher J Cronin, and Meghan M Skira, "Employment effects of healthcare policy: Evidence from the 2007 FDA black box warning on antidepressants," *Journal of Health Economics*, 2020, *73*, 102348. 48

Center on Budget and Policy Priorities, "Chart Book: Social Security Disability Insurance," 2025. 2, 6, 39, 45

Centers for Disease Control and Prevention, National Center for Health Statistics, "National Vital Statistics System, Mortality 1999–2020," CDC WONDER Online Database 2021. Multiple Cause of Death Files, 1999–2020, as compiled from data provided by the 57 vital statistics jurisdictions through the Vital Statistics Cooperative Program. Accessed at `http://wonder.cdc.gov/mcd-icd10.html` on March 27, 2025. 91, 92

Chan, David, Mark Duggan, and Audrey Guo, "Soldiering on: Improving Policies to Benefit America's Veterans," Policy Brief, Stanford Institute for Economic Policy Research February 2021. 50, 26

Chen, Susan and Wilbert van der Klaauw, "The Work Disincentive Effects of the Disability Insurance Program in the 1990s," *Journal of Econometrics*, 2008, *142* (2), 757–784. 45

Chetty, Raj, Nathaniel Hendren, Patrick Kline, and Emmanuel Saez, "Where is the land of Opportunity? The Geography of Intergenerational Mobility in the United States," *The Quarterly Journal of Economics*, November 2014, *129* (4), 1553–1623. 29, 91

_ , _ , _ , _ , **and Nicholas Turner**, "Is the United States still a land of opportunity? Recent trends in intergenerational mobility," *American Economic Review*, 2014, *104* (5), 141–147. 9

Congressional Budget Office, "Policy Options for the Social Security Disability Insurance Program," 2012. 1, 6

County Health Rankings & Roadmaps, "Adult Smoking*," `https://www.countyhealthrankings.org/health-data/community-conditions/health-infrastructure/health-promotion-and-harm-reduction/adult-smoking` 2025. Accessed November 30, 2025. 92

Deshpande, Manasi, "Does Welfare Inhibit Success? The Long-Term Effects of Removing Low-Income Youth from the Disability Rolls," *American Economic Review*, 2016, *106* (11), 3300–3330. 45

_ **and Lee M. Lockwood**, "Beyond Health: Nonhealth Risk and the Value of Disability Insurance," *Econometrica*, 2022, *90* (4), 1781–1810. _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.3982/ECTA19668. 3, 29

_ **and Yue Li**, "Who Is Screened Out? Application Costs and the Targeting of Disability Programs," *American Economic Journal: Economic Policy*, November 2019, *11* (4), 213–248. 4, 29, 40, 46, 19, 20, 32

Dey, Matthew, Elizabeth Weber Handwerker, David S. Piccone Jr, and John Voorheis, "Were wages converging during the 2010s expansion?," *Monthly Labor Review*, June 2022. 32

Dingel, Jonathan I and Brent Neiman, "How many jobs can be done at home?," *Journal of Public Economics*, 2020, *189*, 104235. 9, 40, 44, 93

Fortin, Nicole M., Thomas Lemieux, and Sergio Firpo, "Decomposition Methods in Economics," in Orley Ashenfelter and David Card, eds., *Handbook of Labor Economics*, Vol. 4, Elsevier, 2011, chapter 1, pp. 1–102. 6

French, Eric and Jae Song, "The Effect of Disability Insurance Receipt on Labor Supply," *American Economic Journal: Economic Policy*, 2014, *6* (2), 291–337. 45

Glenn, Karen and Steve Goss, "Disability Experience, Assumptions, and Projections," 2018. 2, 6, 39, 45

Haller, Andreas, Stefan Staubli, and Josef Zweimüller, "Designing Disability Insurance Reforms: Tightening Eligibility Rules or Reducing Benefits?," *Econometrica*, 2024, *92* (1), 79–110. 45

Heim, Bradley, Ithai Lurie, Kathleen J Mullen, and Kosali Simon, "How much do outside options matter? The effect of subsidized health insurance on social security disability insurance benefit receipt," *Journal of Health Economics*, 2021, *76*, 102437. 49, 23

Holford, Theodore R., David T. Levy, Lisa A. McKay, Lauren Clarke, Ben Racine, Rafael Mez, Stephanie Land, Jihyoun Jeon, and Eric J. Feuer, "Patterns of birth cohort-specific smoking histories, 1965-2009," *American Journal of Preventive Medicine*, 2 2014, *46*, e31–e37. 48

Hoynes, Hilary, Nicole Maestas, and Alexander Strand, "Social Security Disability Reform and Implications for Employment," Working Paper NB23-13, National Bureau of Economic Research February 2024. 2, 3, 4, 6, 21, 45

_ , _ , and _ , "Legal Representation in Disability Claims," *Journal of Public Economics*, 2026, *259*, 105661. 47

Isen, Adam, Maya Rossin-Slater, and W. Reed Walker, "Every Breath You Take—Every Dollar You'll Make: The Long-Term Consequences of the Clean Air Act of 1970," *Journal of Political Economy*, 2017, *125* (3), 848–902. 9, 40, 48, 20, 77, 78

Katz, Lawrence F. and David H. Autor, "Changes in the Wage Structure and Earnings Inequality," in Orley Ashenfelter and David Card, eds., *Handbook of Labor Economics*, Vol. 3, Elsevier, 1999, chapter 26, pp. 1463–1555. 6

Kearney, Melissa S, Brendan M Price, and Riley Wilson, "Disability Insurance in the Great Recession: Ease of Access, Program Enrollment, and Local Hysteresis," *AEA Papers and Proceedings*, 2021, *111*, 486–490. 47

Laird, Jessica and Torben Nielsen, "The Effects of Physician Prescribing Behaviors on Prescription Drug Use and Labor Supply: Evidence from Movers in Denmark," *Working paper*, 2016. 48

Liebman, Jeffrey B., "Understanding the Increase in Disability Insurance Benefit Receipt in the United States," *Journal of Economic Perspectives*, May 2015, *29* (2), 123–150. 1, 3, 6, 15, 19, 20, 16

Liu, Siyan and Laura D Quinby, "What factors explain the drop in disability insurance rolls from 2015 to 2019?," *Boston College Center for Retirement Research, Working Paper*, 2023, *7*. 2, 6, 39, 45

Maestas, Nicole, Kathleen J. Mullen, and Alexander Strand, "Disability Insurance and Health Insurance Reform: Evidence from Massachusetts," *American Economic Review*, 2014, *104* (5), 329–335. 49, 23

_ , _ , and _ , "Disability Insurance and the Great Recession," *American Economic Review*, May 2015, *105* (5), 177–182. 6, 30

_ , _ , and _ , "The effect of economic conditions on the disability insurance program: Evidence from the great recession," *Journal of Public Economics*, 2021, *199*, 104410. 1, 6, 30

_ , Kathleen Mullen, and Alexander Strand, "Does Disability Insurance Receipt Discourage Work? Using Examiner Assignment to Estimate Causal Effects of SSDI Receipt," *American Economic Review*, 2013, *103* (5), 1797–1829. 45

_ , Tisamarie B Sherry, and Alexander Strand, "Opioid use among social security disability insurance applicants, 2013–2018," *Journal of Disability Policy Studies*, 2024, *35* (1), 43–53. 2

McVicar, Duncan, Roger Wilkins, and Nicolas R. Ziebarth, "Five Decades of Disability Benefit Policies in Five OECD Countries," in "Work and the Social Safety Net: Labor Activation in Europe and the United States," Oxford University Press, 11 2023. 1

Mohamed, Maiss, Alice Burns, and Juliette Cubanski, "The Connection Between Social Security Disability Benefits and Health Coverage Through Medicaid and Medicare," 2024. 2, 6, 39, 45

Moore, Timothy J., "The Employment Effects of Terminating Disability Benefits," *Journal of Public Economics*, 2015, *124*, 30–43. 45

Ne'eman, Ari and Nicole Maestas, "How Does Medicaid Expansion Impact Income Support Program Participation and Employment for Different Types of People with Disabilities?," Working Paper 31816, National Bureau of Economic Research October 2023. 49, 23, 25

_ **and** _ , "How has COVID-19 impacted disability employment?," *Disability and Health Journal*, 2023, *16* (2), 101429. 6

Oaxaca, Ronald, "Male-Female Wage Differentials in Urban Labor Markets," *International Economic Review*, 1973, *14* (3), 693–709. 2

Reno, Virginia P., "Securing the Future of the Social Security Disability Insurance Program," 2011. 1, 6

Ruffing, Kathy, "Social Security Disability Insurance Still Shrinking," Technical Report, Center on Budget and Policy Priorities January 2020. 2, 39

_ , "Social Security Disability Insurance Still Shrinking," 2023. 2, 6, 39, 45

Schmidt, Lucie, Lara D. Shore-Sheppard, and Tara Watson, "The Impact of the ACA Medicaid Expansion on Disability Program Applications," *American Journal of Health Economics*, September 2020, *6* (4), 444–476. Publisher: The University of Chicago Press. 49, 23, 32, 33

Serena, Benjamin Ly, "The Causal Effect of Scaling up Access to Psychotherapy," September 13 2024. Available at SSRN: `https://ssrn.com/abstract=3797764` or `http://dx.doi.org/10.2139/ssrn.3797764`. 48

Shambaugh, Jay C. and Michael R. Strain, "The Recovery from the Great Recession: A Long, Evolving Expansion," *The ANNALS of the American Academy of Political and Social Science*, 2021, *695* (1), 28–48. 32

Shapiro, Bradley T, "Promoting wellness or waste? evidence from antidepressant advertising," *American Economic Journal: Microeconomics*, 2022, *14* (2), 439–477. 48

Social Security Administration, "Trends in Social Security Disability Insurance," 2019. 2, 6, 39, 45

_ , "The 2023 Annual Report of the Board of Trustees of the Federal Old-Age and Survivors Insurance and Federal Disability Insurance Trust Funds," 2023. 1

_ , "Annual Statistical Report on the Social Security Disability Insurance Program, 2023," 2024. 1, 34, 35, 36

Soni, Aparna, Marguerite E. Burns, Laura Dague, and Kosali I. Simon, "Medicaid Expansion And State Trends In Supplemental Security Income Program Participation," *Health Affairs*, August 2017, *36* (8), 1485–1488. 49, 23

Staiger, Becky, Madeline Helfer, and Jessica Van Parys, "The effect of Medicaid expansion on the take-up of disability benefits by race and ethnicity," *Health Economics*, 2024, *33* (3), 526–540. 49, 23

U.S. Department of Veterans Affairs, "Annual Benefits Report, 2010," 2010. p. 6. 50, 26

_ , "Annual Benefits Report: Compensation, 2023," 2023. p. 73. 50, 26

Vaghul, Kavya and Ben Zipperer, "Historical State and Sub-state Minimum Wages," 2022. Version 1.4.0, https://github.com/benzipperer/historicalminwage/releases/tag/v1.4.0. 9, 90

Wachter, Till Von, Jae Song, and Joyce Manchester, "Trends in Employment and Earnings of Allowed and Rejected Applicants to the Social Security Disability Program," *American Economic Review*, 2011, *101* (7), 3308–3329. 45